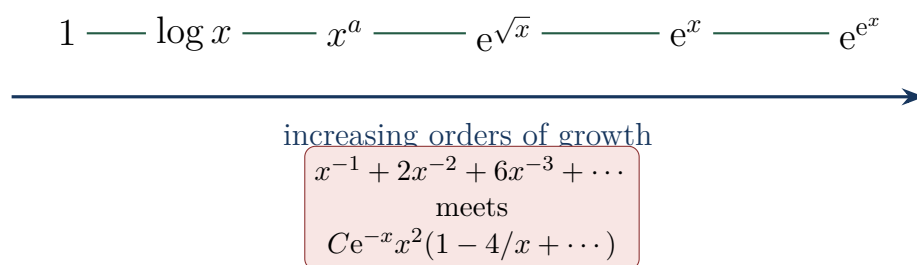


Transseries

A Step-by-Step Tutorial for Mere Mortals

From asymptotic scales and Hahn series to Borel summation, resurgence, and surreal numbers



Prepared as a self-contained mathematical tutorial

August 2026

Preface

Transseries are what happens when ordinary asymptotic series learn to speak the languages of exponentials, logarithms, infinitesimals, infinite quantities, and “beyond-all-orders” effects at the same time. A typical transseries may contain powers of x , iterated logarithms, exponentials such as e^{-x} , and whole asymptotic series multiplying those exponentials:

$$F(x) = \sum_{n \geq 0} \frac{a_n}{x^n} + C e^{-x} x^\beta \sum_{n \geq 0} \frac{b_n}{x^n} + C^2 e^{-2x} x^{2\beta} \sum_{n \geq 0} \frac{c_n}{x^n} + \cdots.$$

This one expression can record ordinary perturbation theory, exponentially small corrections, and the way those corrections interact.

The subject has several overlapping traditions. In algebra and model theory, one studies formal logarithmic-exponential transseries as a large ordered differential field. In asymptotic analysis and mathematical physics, one often studies resurgent transseries whose sectors are connected by Borel-plane singularities and Stokes jumps. In the theory of surreal numbers, transseries appear as canonical infinite expressions evaluated at positive infinite numbers. These traditions share a common grammar but emphasize different questions.

This tutorial starts at the level of a strong second-year undergraduate who has seen calculus, elementary differential equations, and power series. Every major construction is first motivated informally, then stated carefully. Proofs are included when they illuminate the mechanism; technically deep closure theorems are clearly labeled and explained rather than buried under specialist machinery.

Sources and scope

The attached archive supplied with the request contains, among other materials, G. A. Edgar’s expository source *Transseries for Beginners*, his papers on composition and fractional iteration, Dahn–Göring’s notes on exponential-logarithmic terms, work of Kuhlmann–Tressl on LE and EL series, papers of Berarducci–Mantova on surreal numbers and transseries, and an exposition by Aschenbrenner–van den Dries–van der Hoeven on numbers, germs, and transseries. It also contains primers and lecture notes on resurgent transseries and Stokes phenomena. Those sources were supplemented by the original arXiv records, the linked Wikipedia overview as an initial map of terminology, and standard monographs listed in the bibliography [29]. The prose, organization, examples, and pedagogical development here are new; this is not a reproduction of any one source.

How to read this book

The shortest useful path is:

Chapters 1–4 $\longrightarrow$ Chapters 5–10 $\longrightarrow$ Chapters 11–16 $\longrightarrow$ any application chapters you like.

Chapters marked by an “Optional advanced viewpoint” box may be skipped on a first reading. Part V, on Borel summation and resurgence, uses complex variables but begins with concrete real integrals. Part VI explains the wider algebraic landscape without assuming model theory or surreal-number theory.

A promise about notation

The variable x tends to $+\infty$ unless stated otherwise. The relation $f \prec g$ means that $f/g \rightarrow 0$ in an analytic setting, or that the dominant monomial of f is strictly smaller than that of g in a formal setting. We write $f \sim g$ when $f/g \rightarrow 1$, or formally when $f - g \prec g$. The symbol $\asymp$ means “same order of magnitude” and is weaker than $\sim$.

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Part I

Why ordinary asymptotics asks for transseries

Asymptotics is about scale, not convergence

1.1 The question asked by asymptotic analysis

Suppose a function is difficult to evaluate exactly, but we care about it for very large x . A convergent power series is one possible answer, but it is often the wrong kind of answer. Asymptotic analysis asks a more local and more flexible question:

Which simpler terms describe the function in decreasing order of importance as $x \rightarrow +\infty$?

The phrase “decreasing order of importance” is the heart of the subject. The sequence

$$1, \quad x^{-1}, \quad x^{-2}, \quad x^{-3}, \dots$$

is an **asymptotic scale** because each term is negligible compared with the preceding one:

$$x^{-(n+1)} = o(x^{-n}).$$

More generally, a sequence (ϕ_n) is an asymptotic scale if

$$\phi_{n+1}(x) = o(\phi_n(x)) \quad (x \rightarrow +\infty).$$

Definition 1.1 (Poincaré asymptotic expansion). Let $(\phi_n)_{n \geq 0}$ be an asymptotic scale. We write

$$f(x) \sim_{\text{as}} \sum_{n=0}^{\infty} a_n \phi_n(x)$$

if, for every fixed $N \geq 1$,

$$f(x) - \sum_{n=0}^{N-1} a_n \phi_n(x) = o(\phi_{N-1}(x)).$$

A slightly stronger and common version requires the remainder to be $O(\phi_N)$.

This definition deliberately says nothing about convergence of the infinite sum. It only says that each fixed truncation becomes more accurate in a precise relative sense.

Worked example: the geometric series viewed from infinity

For $x > 1$,

$$\frac{x}{x-1} = \frac{1}{1-x^{-1}} = 1 + x^{-1} + x^{-2} + \dots$$

Here the series actually converges. Its N -term remainder is

$$\frac{x^{-N}}{1-x^{-1}} = O(x^{-N}),$$

so it is also an asymptotic expansion. Convergence is a bonus, not part of the definition.

1.2 A divergent series can be an excellent approximation

Consider Euler's factorial series

$$\tilde{F}(x) = \sum_{n=0}^{\infty} (-1)^n n! x^{-n-1}.$$

For every fixed x , the terms eventually grow because

$$\frac{(n+1)!x^{-n-2}}{n!x^{-n-1}} = \frac{n+1}{x}.$$

Thus the series diverges for every finite x . Nevertheless, it is the asymptotic expansion of the perfectly ordinary function

$$F(x) = \int_0^{\infty} \frac{e^{-xp}}{1+p} dp = -e^x \operatorname{Ei}(-x), \quad x > 0.$$

Repeated integration by parts gives, for every N ,

$$F(x) = \sum_{n=0}^{N-1} (-1)^n n! x^{-n-1} + (-1)^N N! \int_0^{\infty} \frac{e^{-xp}}{(1+p)^{N+1}} dp.$$

The final integral is $O(x^{-N-1})$, so the divergent series is asymptotic to F .

Intuition

A convergent series is designed to be summed all the way to infinity. A divergent asymptotic series is designed to be stopped. Its early terms improve the approximation; its late terms eventually make it worse. The location of the smallest term is usually the best stopping point.

If the n th term behaves like $n!/x^{n+1}$, then it decreases until roughly $n \approx x$ and increases afterward. Truncating near $n = x$ leaves an error on the order of e^{-x} , which is smaller than every power x^{-N} . This observation points directly toward transseries.

1.3 Beyond all algebraic orders

The elementary limit

$$e^{-x} = o(x^{-N}) \quad \text{for every fixed } N$$

means that the exponential e^{-x} is invisible to the entire power scale

$$1, x^{-1}, x^{-2}, \dots$$

Consequently, the functions 0 and e^{-x} have the same asymptotic power series: all coefficients are zero. Likewise,

$$\frac{1}{x} + e^{-x} \quad \text{and} \quad \frac{1}{x} + 17e^{-x}$$

have the same expansion in inverse powers of x .

A power series therefore cannot distinguish functions that differ by an exponentially small amount. A transseries repairs this loss of information by extending the scale:

$$1 \succ x^{-1} \succ x^{-2} \succ \dots \succ e^{-x} \succ x^{-1}e^{-x} \succ \dots \succ e^{-2x} \succ \dots$$

The notation $A \succ B$ means that A is asymptotically much larger than B .

Worked example: a boundary layer that ordinary perturbation misses

Let $0 < \varepsilon \ll 1$ and solve

$$\varepsilon y'(t) + y(t) = 1, \quad y(0) = 0.$$

The exact solution is

$$y(t) = 1 - e^{-t/\varepsilon}.$$

For every fixed $t > 0$, the exponential is smaller than every power of ε :

$$e^{-t/\varepsilon} = o(\varepsilon^N) \quad \text{for all } N.$$

A power-series ansatz $y \sim y_0 + \varepsilon y_1 + \dots$ produces only $y \sim 1$. The missing term is not a minor correction near $t = 0$; it is the entire mechanism that enforces the initial condition. In the variable $x = 1/\varepsilon$, the complete expression is the two-term transseries

$$y(t) = 1 - e^{-tx}.$$

1.4 Optimal truncation and the first glimpse of resurgence

Suppose

$$a_n \sim C A^{-n} \Gamma(n + \beta).$$

Then the terms $a_n x^{-n}$ are smallest near $n \approx Ax$. Stirling's formula suggests that the least term is on the scale

$$x^{\beta-1/2} e^{-Ax}.$$

Thus factorial divergence in the perturbative coefficients predicts an exponentially small scale. In resurgent problems this is not an accident: the large-order growth of one sector encodes the existence and structure of another sector. We will make this statement concrete in Part V.

Checkpoint

At this stage you should be comfortable with four distinct ideas:

1. an asymptotic expansion is a hierarchy of remainders, not necessarily a convergent sum;

2. a divergent series may approximate a function extremely well when truncated;
3. exponentials such as e^{-x} are smaller than every inverse power;
4. a richer expansion must include several asymptotic scales at once.

The growth ladder

2.1 Comparing familiar functions

For large positive x we have the chain

$$\log \log x \prec \log x \prec x^a \prec e^{x^b} \prec e^x \prec e^{e^x}$$

whenever $a, b > 0$ and $0 < b < 1$. The comparison is multiplicative. For example,

$$\frac{\log x}{x^a} \rightarrow 0, \quad \frac{x^a}{e^{x^b}} \rightarrow 0, \quad \frac{e^x}{e^{e^x}} \rightarrow 0.$$

This is not merely a list of limits; it behaves like an ordered multiplicative group. Products, reciprocals, and real powers remain meaningful:

$$\frac{x^{\sqrt{2}}e^{-x}}{(\log x)^3}, \quad x^\pi e^{x^{1/2}-2x^{1/3}}, \quad \exp\left(-\frac{x}{\log x}\right).$$

Such building blocks are called **transmonomials**.

2.2 Three comparison relations

For positive quantities f and g near infinity:

$$\begin{aligned} f \prec g &\iff f/g \rightarrow 0, \\ f \asymp g &\iff f/g \rightarrow c \in (0, \infty), \\ f \sim g &\iff f/g \rightarrow 1. \end{aligned}$$

The first relation compares orders of magnitude, the second says “same monomial scale,” and the third says “same leading term.” For instance,

$$3x^2 + \log x \asymp x^2, \quad 3x^2 + \log x \sim 3x^2, \quad \log x \prec x^2.$$

2.3 Why the resulting field is non-Archimedean

In the real numbers, no positive number is larger than every integer. In a formal field containing an infinite symbol x , we impose

$$x > n \quad \text{for every } n \in \mathbb{N}.$$

Then x^{-1} is a positive infinitesimal:

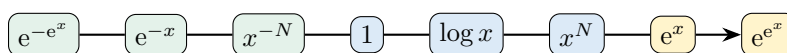
$$0 < x^{-1} < \frac{1}{n} \quad \text{for every } n \in \mathbb{N}.$$

An ordered field with such elements is **non-Archimedean**. This feature is not pathology; it is exactly what allows the field to store asymptotic size algebraically.

Core idea

A transseries does not first receive a numerical value and then get compared. Its ordering is read from its largest monomial. The asymptotic hierarchy is built into the syntax.

2.4 A visual growth map



larger monomials to the right
reciprocals reflect the picture about 1

The actual transmonomial group is vastly denser than this picture. Between x and e^x lie $e^{\sqrt{x}}$, $e^{x/\log x}$, $e^{x^{0.999}}$, and infinitely many other scales. Between e^{-x} and e^{-2x} lie $e^{-x-x^{1/2}}$, $x^{-100}e^{-x}$, and so on.

2.5 Comparability and oscillation

The classical logarithmic-exponential world is designed for eventually non-oscillatory growth. The functions x , $\log x$, and e^x can be consistently ranked. The function $\sin x$ cannot: it changes sign infinitely often and is neither eventually larger nor eventually smaller than zero.

This explains a basic limitation. Standard real LE-transseries do not directly contain a monomial behaving like $\sin x$. Complex exponentials e^{ix} can be adjoined in other frameworks, but then the simple total ordering is lost. Transseries are exceptionally powerful for regular, non-oscillatory asymptotics; they are not intended to replace Fourier analysis.

Your first transseries

3.1 An informal definition

A transseries is a formal sum

$$T = \sum_{\mathfrak{m} \in S} a_{\mathfrak{m}} \mathfrak{m}, \quad a_{\mathfrak{m}} \in \mathbb{R},$$

where the symbols $\mathfrak{m}$ are transmonomials and the support S is arranged so that there is always a largest remaining monomial. Typical examples are

$$T_1 = x^3 - 2x + 7 + x^{-1} + 2x^{-2} + 6x^{-3} + \cdots,$$

$$T_2 = \log x - \log \log x + \frac{\log \log x}{\log x} + \cdots,$$

$$T_3 = 1 + \sum_{n \geq 0} a_n x^{-n-1} + C e^{-x} x^2 \sum_{n \geq 0} b_n x^{-n},$$

$$T_4 = e^{x + \sqrt{x} + x^{1/3} + \cdots} + x^\pi - \log x + e^{-x}.$$

The infinite dots are allowed only when the set of monomials is well organized. Formal generalized series are not arbitrary bags of terms.

3.2 Dominant term, magnitude, and sign

If

$$T = 3e^x - x^{100} + 7 + e^{-x} + \cdots,$$

then the dominant monomial is e^x , the dominant term is $3e^x$, and T is positive because the leading coefficient is positive. We write

$$\text{mag}(T) = e^x, \quad \text{dom}(T) = 3e^x.$$

The lower terms cannot overturn the sign of the leading term, just as $3x^5 - 10^{100}x^4$ is positive for sufficiently large x .

Every nonzero transseries has a multiplicative normal form

$$T = a \mathfrak{m}(1 + \varepsilon), \quad a \in \mathbb{R}^\times, \quad \mathfrak{m} = \text{mag}(T), \quad \varepsilon \prec 1.$$

This one decomposition drives inversion, logarithms, real powers, and many recursive calculations.

3.3 Examples of legal and illegal supports

At $x \rightarrow +\infty$, the formal series

$$1 + x^{-1} + x^{-2} + \dots$$

is legal because its monomials decrease. The series

$$1 + x + x^2 + x^3 + \dots$$

is not a classical transseries support: there is no largest monomial, so the expression has no dominant term.

Likewise,

$$\log x + \log \log x + \log \log \log x + \dots$$

uses unbounded logarithmic depth. It does not belong to the classical finite-depth LE field, although larger systems such as omega-series and hyperseries are designed to include related expressions.

A third non-example is

$$x + e^x + e^{e^x} + e^{e^{e^x}} + \dots,$$

which has unbounded exponential height. Again, larger transserial fields can go beyond the classical boundary.

Common pitfall

“Every term gets smaller” is not by itself sufficient. One also needs a support condition that makes products and infinite sums well defined. The precise condition is reverse well-ordering, introduced in Chapter 6.

3.4 Four elementary computations

Worked example: a geometric inverse

Since $x^{-1} \prec 1$,

$$(1 - x^{-1})^{-1} = 1 + x^{-1} + x^{-2} + \dots.$$

Therefore

$$(x - 1)^{-1} = x^{-1} + x^{-2} + x^{-3} + \dots.$$

Multiplication is formal, and every coefficient receives only finitely many contributions.

Worked example: the inverse of $e^x + x$

Factor out the dominant monomial:

$$e^x + x = e^x(1 + xe^{-x}).$$

The correction xe^{-x} is small, so

$$(e^x + x)^{-1} = e^{-x} \sum_{n \geq 0} (-1)^n x^n e^{-nx} = \sum_{n \geq 0} (-1)^n x^n e^{-(n+1)x}.$$

The ordinary geometric series has been transplanted to an exponential scale.

Worked example: the logarithm of $\sinh x$

For large positive x ,

$$\sinh x = \frac{e^x}{2}(1 - e^{-2x}).$$

Hence

$$\log(\sinh x) = x - \log 2 + \log(1 - e^{-2x}) = x - \log 2 - \sum_{n \geq 1} \frac{e^{-2nx}}{n}.$$

An elementary function becomes an infinite exponential transseries.

Worked example: an algebraic equation with exponential sectors

Solve

$$y^2 - y = e^{-x}.$$

The two roots are

$$y_{\pm} = \frac{1 \pm \sqrt{1 + 4e^{-x}}}{2}.$$

Expanding the square root gives

$$\begin{aligned} y_+ &= 1 + e^{-x} - e^{-2x} + 2e^{-3x} - 5e^{-4x} + \dots, \\ y_- &= -e^{-x} + e^{-2x} - 2e^{-3x} + 5e^{-4x} - \dots. \end{aligned}$$

The coefficients are signed Catalan numbers. Even an algebraic equation can naturally produce a purely exponential transseries.

3.5 The decomposition $T = L + c + S$

Every real transseries can be split uniquely as

$$T = L + c + S,$$

where L is **purely large** (all its monomials are $\succ 1$), $c \in \mathbb{R}$, and S is **small** (all its monomials are $\prec 1$). For example,

$$x + \sqrt{x} - 3 + \frac{\log x}{x} + e^{-x} = \underbrace{x + \sqrt{x}}_L \underbrace{- 3}_c + \underbrace{\frac{\log x}{x} + e^{-x}}_S.$$

Then exponentiation is defined by

$$e^T = e^L e^c \sum_{n \geq 0} \frac{S^n}{n!}.$$

This formula illustrates the recursive nature of the subject: the new monomial e^L is multiplied by a generalized power series in the small quantity S .

What the word “transseries” can mean

4.1 One word, several closely related objects

The literature uses “transseries” for a family of constructions rather than one universally fixed set. The differences matter, but they should not obscure the common idea.

4.1.1 Formal logarithmic-exponential transseries

These are generalized series built recursively from real constants, an infinite variable x , real powers, exponentials, logarithms, and admissible infinite sums. They form an ordered differential field usually denoted $\mathbb{T}$. This is the main algebraic object in Parts II–IV. Edgar’s beginner exposition [17], the constructions of van den Dries–Macintyre–Marker [11, 12], and van der Hoeven’s monograph [23] are central references.

4.1.2 Grid-based and well-based transseries

Grid-based transseries have supports contained in finitely generated multiplicative grids. They are convenient for algorithms and explicit calculations. Well-based transseries allow arbitrary reverse well-ordered supports and are more flexible. Most formulas look the same, but some closure and convergence arguments require separate proofs. We compare them in Chapter 7.

4.1.3 Écalle’s analyzable and resurgent transseries

In resurgence theory a transseries often has the sector form

$$\Phi(x; \sigma) = \sum_{k \geq 0} \sigma^k e^{-kAx} x^{k\beta} \Phi_k(x), \quad \Phi_k(x) = \sum_{n \geq 0} a_{k,n} x^{-n}.$$

The focus is not only formal algebra. One studies Borel transforms, analytic continuation, Stokes automorphisms, and the way different sectors determine one another. This tradition descends from Écalle’s work [16] and is developed in modern expositions such as Costin [10], Dorigoni [15], and Aniceto–Başar–Schiappa [1].

4.1.4 Transseries as germs or surreal functions

A **germ at infinity** identifies two functions if they agree for all sufficiently large x . Hardy fields organize non-oscillating germs. In another direction, one can evaluate suitable formal series at

positive infinite surreal numbers. The H -field viewpoint creates a bridge among germs, transseries, and surreal numbers [2, 3, 5, 6].

4.2 Formal object versus represented function

It is essential to separate three statements:

1. T is a legal formal transseries.
2. A function f has T as an asymptotic expansion.
3. A summation procedure assigns a specific analytic function to T .

The first is algebraic. The second is an estimate about remainders. The third requires extra analytic information and may depend on a direction in the complex plane.

Common pitfall

A formal transseries is not automatically a convergent numerical expression. Manipulating it can be perfectly rigorous even when no ordinary infinite sum exists. This is no stranger than manipulating a formal power series in algebra.

4.3 A useful working convention

Throughout the formal chapters we use real, well-based logarithmic-exponential transseries at $x \rightarrow +\infty$. When a proof is easiest in the grid-based setting, we say so. In the analytic chapters, x may be complex and the word “sector” refers to an angular region where a Borel-Laplace sum is defined.

4.4 Historical orientation

The intellectual roots include du Bois-Reymond’s orders of infinity, Hardy’s logarithmico-exponential functions, Hahn’s generalized series, and Borel’s theory of divergent series. Modern transseries were developed independently in the 1980s by Dahn and Göring in connection with exponential fields and by Écalte in his work on analyzable functions and the Dulac problem. Later developments connected the subject with computer algebra, o-minimality, differential algebra, model theory, resurgence, and surreal numbers; see [13, 16, 12, 23, 2].

Checkpoint

You are now ready for the formal core. The key ingredients will be:

1. an ordered multiplicative group of monomials;
2. generalized series whose supports are reverse well ordered;
3. recursive closure under exponentials and logarithms;
4. operations controlled by dominant monomials and small remainders.

Part II

Generalized series and the formal construction

Ordered monomial groups

5.1 Why monomials should form a group

Ordinary Laurent series use monomials x^n with $n \in \mathbb{Z}$. They multiply according to

$$x^m x^n = x^{m+n}, \quad (x^n)^{-1} = x^{-n}.$$

The exponents form the additive group $\mathbb{Z}$, while the monomials form a multiplicative group. Generalized series keep this pattern but replace $\mathbb{Z}$ by a much richer ordered group.

Definition 5.1 (ordered abelian group). An **ordered abelian group** $(\mathfrak{G}, \cdot, 1, \prec)$ is an abelian group together with a total order preserved by multiplication:

$$\mathfrak{m} \prec \mathfrak{n} \implies \mathfrak{m}g \prec \mathfrak{n}g \quad (g \in \mathfrak{G}).$$

We think of elements of $\mathfrak{G}$ as positive monomials. The notation $\mathfrak{m} \prec \mathfrak{n}$ means that $\mathfrak{m}$ is asymptotically smaller than $\mathfrak{n}$.

The simplest example is

$$\mathfrak{G}_0 = x^{\mathbb{R}} := \{x^a : a \in \mathbb{R}\},$$

ordered by $x^a \prec x^b$ exactly when $a < b$. This already allows generalized power series such as

$$3x^{\sqrt{2}} - 7x^{\pi} + x^{-1/3} + \dots$$

5.2 Multiplicative notation and logarithmic coordinates

Ordered abelian groups are often written additively. If Γ is additive, the associated monomial group may be written

$$t^{\Gamma} = \{t^{\gamma} : \gamma \in \Gamma\}, \quad t^{\alpha} t^{\beta} = t^{\alpha+\beta}.$$

Transseries are more naturally written multiplicatively because their monomials look like products and exponentials. Formally, however, taking a logarithm turns the monomial group into an additive ordered group:

$$\log(\mathfrak{m}\mathfrak{n}) = \log \mathfrak{m} + \log \mathfrak{n}.$$

For instance,

$$\mathfrak{m} = x^a e^L \implies \log \mathfrak{m} = a \log x + L.$$

This observation is what makes exponentiation fit recursively into the construction.

5.3 Coefficients are not part of the monomial

A term has the form $a\mathbf{m}$ with $a \in \mathbb{R}^\times$ and $\mathbf{m} \in \mathfrak{G}$. The coefficient a affects sign and leading constant but not order of magnitude. Thus

$$7x^3 \asymp -10^{100}x^3, \quad x^3 \succ x^2,$$

although $7x^3$ and $-10^{100}x^3$ have opposite signs.

The separation of coefficient and monomial gives a unique normal form for each nonzero term. This uniqueness is essential when like terms are collected.

5.4 A lexicographic example

Consider monomials

$$\mathbf{m} = x^a e^L,$$

where L is purely large. Compare two of them by first comparing the exponential parts:

$$x^{a_1} e^{L_1} \prec x^{a_2} e^{L_2}$$

when either $L_1 < L_2$, or $L_1 = L_2$ and $a_1 < a_2$. This is a lexicographic order. It expresses the elementary fact that an exponential difference dominates every power:

$$L_2 - L_1 \succ 1 \implies \frac{x^{a_1} e^{L_1}}{x^{a_2} e^{L_2}} = x^{a_1 - a_2} e^{-(L_2 - L_1)} \prec 1.$$

Worked example: sorting a mixed collection of monomials

Arrange the following from largest to smallest:

$$e^{x+\sqrt{x}}, \quad x^{100}e^x, \quad x^{-3}e^x, \quad e^{\sqrt{x}}, \quad x^\pi, \quad e^{-x}.$$

First compare exponential exponents:

$$x + \sqrt{x} > x > \sqrt{x} > 0 > -x.$$

Within the equal exponent x , compare powers of x . The result is

$$e^{x+\sqrt{x}} \succ x^{100}e^x \succ x^{-3}e^x \succ e^{\sqrt{x}} \succ x^\pi \succ e^{-x}.$$

5.5 Monomial logarithmic derivatives

For a nonzero differentiable monomial $\mathbf{m}$, define

$$\mathbf{m}^\dagger := \frac{\mathbf{m}'}{\mathbf{m}}.$$

This is the **logarithmic derivative**. If $\mathbf{m} = x^a e^L$, then

$$\mathbf{m}^\dagger = \frac{a}{x} + L'.$$

The logarithmic derivative is often simpler than the derivative itself and is a central tool for asymptotic integration and differential equations.

Example 5.2. For

$$\mathfrak{m} = x^3 \exp(-x^2 + \sqrt{x}),$$

we have

$$\mathfrak{m}^\dagger = \frac{3}{x} - 2x + \frac{1}{2\sqrt{x}}, \quad \mathfrak{m}' = \left(-2x + \frac{1}{2\sqrt{x}} + \frac{3}{x}\right) \mathfrak{m}.$$

The dominant factor in the derivative is $-2x\mathfrak{m}$.

5.6 The role of finite height

A classical transmonomial is built using only finitely many nested exponentials and logarithms. Each individual monomial therefore has finite complexity. A transseries may have infinitely many terms, but there is a finite bound on the exponential height and logarithmic depth appearing in a given classical LE-transseries.

This restriction makes the recursive construction manageable. It also explains why an infinite exponential tower is not an element of the classical field. Later systems of hyperseries relax this boundary.

Hahn series and well-based support

6.1 Generalized series

Fix an ordered abelian monomial group $\mathfrak{G}$. A formal generalized series is an expression

$$F = \sum_{\mathfrak{m} \in \mathfrak{G}} a_{\mathfrak{m}} \mathfrak{m},$$

where only a selected set of coefficients is nonzero. The support is

$$\text{supp } F = \{\mathfrak{m} \in \mathfrak{G} : a_{\mathfrak{m}} \neq 0\}.$$

If arbitrary supports were allowed, multiplication could require infinite, undefined coefficient sums. Hahn's insight was to impose an order condition on the support.

Definition 6.1 (reverse well-ordered support). A subset $S \subseteq \mathfrak{G}$ is **reverse well ordered** if every nonempty subset of S has a largest element. Equivalently, there is no infinite strictly increasing chain

$$\mathfrak{m}_1 \prec \mathfrak{m}_2 \prec \mathfrak{m}_3 \prec \cdots$$

inside S .

Many authors reverse the group order and simply say “well ordered.” In transseries it is natural to write terms from largest to smallest, so “reverse well ordered” is the transparent phrase.

Example 6.2. The set

$$\{1, x^{-1}, x^{-2}, \dots\}$$

is reverse well ordered. Its largest member is 1, and every tail has a largest member. By contrast,

$$\{1, x, x^2, \dots\}$$

has no largest member and is not reverse well ordered.

6.2 Hahn series

Definition 6.3 (Hahn field notation). The set of all real generalized series with reverse well-ordered support in $\mathfrak{G}$ is denoted

$$\mathbb{R}((\mathfrak{G}))$$

or, in transseries typography, $\mathbb{R}[[\mathfrak{G}]]$.

A nonzero Hahn series automatically has a largest monomial, hence a dominant term and a sign. This turns the formal series field into an ordered field.

6.3 Addition

Addition is coefficientwise:

$$\left(\sum_{\mathfrak{m}} a_{\mathfrak{m}} \mathfrak{m} \right) + \left(\sum_{\mathfrak{m}} b_{\mathfrak{m}} \mathfrak{m} \right) = \sum_{\mathfrak{m}} (a_{\mathfrak{m}} + b_{\mathfrak{m}}) \mathfrak{m}.$$

The support of the sum is contained in the union of the two supports. A finite union of reverse well-ordered sets is reverse well ordered, so addition is safe.

6.4 Multiplication and the Hahn–Neumann mechanism

Formally distribute:

$$\left(\sum_{\mathfrak{m}} a_{\mathfrak{m}} \mathfrak{m} \right) \left(\sum_{\mathfrak{n}} b_{\mathfrak{n}} \mathfrak{n} \right) = \sum_{\mathfrak{g} \in \mathfrak{G}} \left(\sum_{\mathfrak{m}\mathfrak{n}=\mathfrak{g}} a_{\mathfrak{m}} b_{\mathfrak{n}} \right) \mathfrak{g}.$$

Two facts are required:

1. the product support is reverse well ordered;
2. for each fixed $\mathfrak{g}$, only finitely many pairs $(\mathfrak{m}, \mathfrak{n})$ contribute.

A classical lemma of Hahn and Neumann supplies these facts under the support hypotheses. This is the hidden combinatorial engine behind generalized-series multiplication.

Worked example: a support of order type ω^2

Let

$$U = \sum_{j \geq 1} j e^{-jx}, \quad V = \sum_{k \geq 0} x^{-k}.$$

Then

$$UV = \sum_{j \geq 1} \sum_{k \geq 0} j x^{-k} e^{-jx}.$$

For fixed j , the powers $x^{-k} e^{-jx}$ descend as k increases. After all terms with $j = 1$ come those with $j = 2$, and so on. The support has the nested ordering pattern

$$\omega + \omega + \omega + \cdots = \omega^2.$$

Transfinite order types appear naturally even though every displayed calculation is elementary.

6.5 Summable families

To define an infinite sum of Hahn series, one needs a stronger notion than termwise convergence.

Definition 6.4 (summable family). A family $(F_i)_{i \in I}$ of Hahn series is **summable** if

1. the union $\bigcup_i \text{supp } F_i$ is reverse well ordered;
2. each monomial belongs to the supports of only finitely many F_i .

Then $\sum_i F_i$ is defined by finite coefficientwise summation at every monomial.

This notion explains why the geometric series

$$\sum_{n \geq 0} \varepsilon^n$$

is legal whenever $\varepsilon \prec 1$: the supports move downward, and each target monomial receives finitely many contributions under the standard hypotheses.

Common pitfall

Formal summability is not numerical convergence. A summable family may define a valid Hahn series even when substituting a finite real number for x makes the numerical series diverge.

6.6 The field theorem

Theorem 6.5 (Hahn). *If K is a field and $\mathfrak{G}$ is an ordered abelian group, then the generalized series $K((\mathfrak{G}))$ with reverse well-ordered support form a field.*

The nontrivial step is inversion. Let

$$F = a\mathfrak{m}(1 + \varepsilon), \quad \varepsilon \prec 1.$$

Then

$$F^{-1} = a^{-1}\mathfrak{m}^{-1} \sum_{n \geq 0} (-\varepsilon)^n.$$

The generalized geometric family is summable, so the inverse exists formally.

Proof idea. The leading term $a\mathfrak{m}$ can be inverted in the coefficient field and monomial group. The remainder ε is small. Every power ε^n has monomials strictly below those in ε^{n-1} in the relevant multiplicative sense. Neumann’s lemma shows that the union of supports remains well based and that each coefficient is a finite sum. Formal multiplication then gives

$$(1 + \varepsilon) \sum_{n \geq 0} (-\varepsilon)^n = 1.$$

□

6.7 Truncations

For a cutoff monomial $\mathfrak{g}$, define the truncation

$$F_{\succeq \mathfrak{g}} := \sum_{\mathfrak{m} \succeq \mathfrak{g}} a_{\mathfrak{m}}\mathfrak{m}.$$

A field of generalized series is **truncation closed** if these initial segments remain in the field. Classical transseries fields are constructed to have strong truncation properties. Truncation is the formal analogue of computing “all terms down to a requested size.”

Core idea

A transseries computation is exact above a chosen cutoff. One need not compute an infinite expression all at once. The support order guarantees that only finitely much algebra influences any fixed leading segment.

Grid-based versus well-based series

7.1 Ratio sets

A grid-based series begins with finitely many small monomials

$$\boldsymbol{\mu} = (\mu_1, \dots, \mu_r), \quad \mu_i \prec 1.$$

For an integer vector $\mathbf{k} = (k_1, \dots, k_r)$ write

$$\boldsymbol{\mu}^{\mathbf{k}} = \mu_1^{k_1} \cdots \mu_r^{k_r}.$$

Given a lower-bound vector $\mathbf{m} \in \mathbb{Z}^r$, the associated grid is

$$\mathfrak{J}^{\boldsymbol{\mu}, \mathbf{m}} = \left\{ \boldsymbol{\mu}^{\mathbf{k}} : \mathbf{k} \geq \mathbf{m} \right\}$$

with coordinatewise inequality.

Example 7.1. Take $\mu_1 = x^{-1}$ and $\mu_2 = e^{-x}$. The grid contains

$$x^{-k} e^{-jx} \quad (k, j \geq 0).$$

A two-scale transseries such as

$$\sum_{j \geq 0} e^{-jx} \sum_{k \geq 0} a_{j,k} x^{-k}$$

is grid based.

7.2 Why finite grids help computation

A finite ratio set turns support bookkeeping into integer-vector bookkeeping. Multiplication adds exponent vectors. To compute down to a cutoff, algorithms can enumerate a finite region of $\mathbb{Z}^r$. Dickson's lemma—every subset of $\mathbb{N}^r$ has finitely many minimal elements—underlies many finiteness arguments.

Grid-based transseries are therefore natural in computer algebra. Van der Hoeven's approach develops effective operations and Newton methods in this setting [23].

7.3 Why well-based series are more general

Not every reverse well-ordered support lies in a finitely generated grid. For example, a family of monomials may involve infinitely many independent scales. Well-based series permit such supports as long as the order condition holds.

The tradeoff is familiar:

	Grid based	Well based
Support	contained in a finite-rank grid	arbitrary reverse well-ordered set
Algorithms	especially concrete	more abstract
Closure proofs	often combinatorial in $\mathbb{Z}^r$	rely on general support lemmas
Generality	smaller class	larger class

7.4 Witnesses and ratio sets

In grid-based work, a finite ratio set can **witness** a comparison such as $A \prec B$ by expressing every monomial in A/B as a positive product of the chosen small ratios. This gives an effective certificate that a remainder is small.

For example, with $\mu = (x^{-1}, e^{-x})$, the transseries

$$S = 3x^{-2} + xe^{-x} + e^{-2x}$$

is small because every monomial is generated by products of x^{-1} and e^{-x} with exponents in a suitable grid. The positive power x in xe^{-x} does not make the term large; the exponential factor wins.

7.5 Which version will we use?

We state the basic algebra in well-based language because it reveals the general idea. Most explicit examples live in a small grid and may be computed as if one were manipulating multivariate formal power series. When composition, convergence, or fractional iteration has version-dependent subtleties, we mention them.

Constructing logarithmic-exponential transseries

8.1 Stage zero: generalized powers

Begin with the monomial group

$$\mathfrak{G}_0 = x^{\mathbb{R}}$$

and the Hahn field

$$\mathbb{T}_0 = \mathbb{R}((x^{\mathbb{R}})).$$

Elements of $\mathbb{T}_0$ are generalized power series

$$\sum_{a \in S} c_a x^a$$

with reverse well-ordered exponent support. This stage already contains Laurent series, Puiseux series, and real powers.

Every $F \in \mathbb{T}_0$ decomposes as

$$F = L + c + S,$$

where L contains powers x^a with $a > 0$, c is constant, and S contains powers with $a < 0$.

8.2 Adjoining exponentials

Assume a field $\mathbb{T}_n$ has been built. Let $\mathbb{T}_n^{\text{pl}}$ denote its purely large elements. Form new monomials

$$\mathfrak{G}_{n+1} = \left\{ x^b e^L : b \in \mathbb{R}, L \in \mathbb{T}_n^{\text{pl}} \right\}.$$

Then form the next Hahn field

$$\mathbb{T}_{n+1} = \mathbb{R}((\mathfrak{G}_{n+1})).$$

The **exponential height** counts how many such stages are needed.

Example 8.1. The monomial $e^{x^2 + \sqrt{x}}$ has height one over generalized powers. The monomial

$$\exp(e^x + x^{1/2})$$

has height two because its exponent already contains e^x .

The log-free field is the union

$$\mathbb{T}_\bullet = \bigcup_{n \geq 0} \mathbb{T}_n,$$

with compatible embeddings between stages. The full construction takes care that a series represented at one stage keeps the same meaning at later stages.

8.3 Defining the exponential of an arbitrary transseries

Let

$$T = L + c + S$$

with L purely large, $c \in \mathbb{R}$, and $S \prec 1$. Define

$$\exp(T) = e^L e^c \sum_{n \geq 0} \frac{S^n}{n!}.$$

The factor e^L is a monomial introduced at the next height, while the Taylor series in S is a summable Hahn family.

Worked example: expanding an exponential

Let

$$T = x + \log x + x^{-1} + 2e^{-x}.$$

Treat x as purely large, $\log x$ as part of the monomial exponent after logarithms are admitted, and

$$S = x^{-1} + 2e^{-x}$$

as small. Then

$$e^T = x e^x \exp(x^{-1} + 2e^{-x}).$$

To a few orders,

$$\exp(x^{-1} + 2e^{-x}) = 1 + x^{-1} + \frac{1}{2}x^{-2} + 2e^{-x} + 2x^{-1}e^{-x} + 2e^{-2x} + \dots$$

The ordering decides which of these terms should be displayed first.

8.4 Adjoining logarithms

A positive transseries has normal form

$$T = a \mathfrak{m}(1 + \varepsilon), \quad a > 0, \quad \varepsilon \prec 1.$$

Define

$$\log T = \log a + \log \mathfrak{m} + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \varepsilon^n.$$

For a monomial $\mathfrak{m} = x^b e^L$,

$$\log \mathfrak{m} = b \log x + L.$$

To make $\log x$, $\log \log x$, and further iterates available, one may substitute $\log_M x$ for x in the log-free field, where

$$\log_0 x = x, \quad \log_{M+1} x = \log(\log_M x).$$

The classical LE field is a union over finite height and finite logarithmic depth.

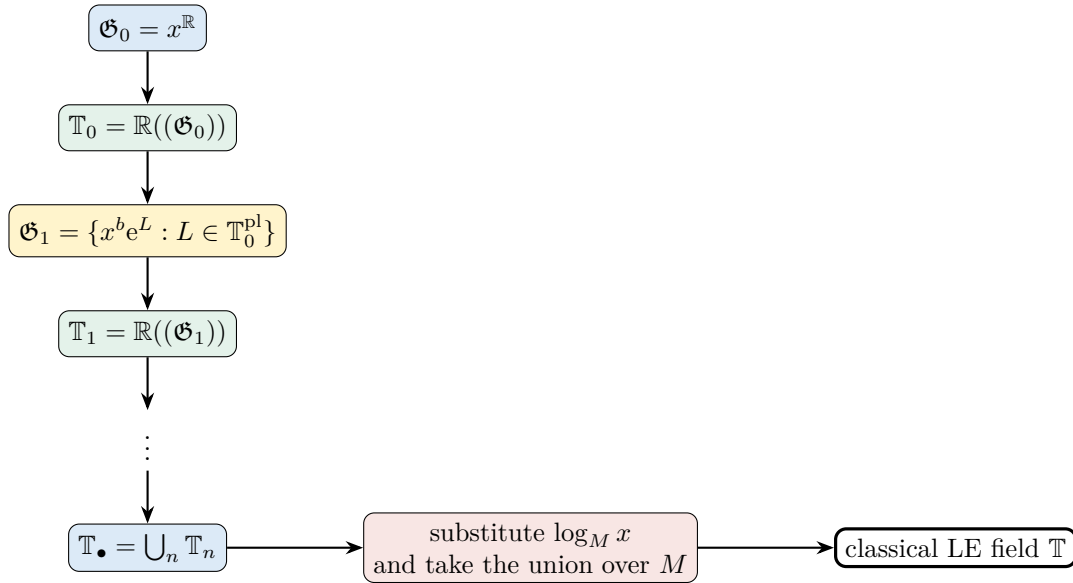
8.5 Why the recursive definition is not circular

At first sight, “a monomial is the exponential of a transseries, and a transseries is a sum of monomials” sounds circular. The height filtration resolves the issue:

1. stage 0 monomials are known;
2. exponentials at stage $n + 1$ use purely large exponents from earlier stages;
3. Hahn sums are then formed from the newly known monomials.

Every individual object appears at a finite stage, so the recursion is well founded.

8.6 A schematic of the construction



Optional advanced viewpoint

There are several equivalent or closely related constructions: logarithmic-exponential series, exponential-logarithmic series, grid-based transseries, well-based transseries, and fields built inside the surreal numbers. Their exact closure procedures differ. For a first course, the height-and-depth picture above captures the operational core, while the references supply the set-theoretic details [12, 23, 26, 28].

Dominance and normal forms

9.1 The dominant monomial

Let

$$T = \sum_{\mathbf{m} \in S} a_{\mathbf{m}} \mathbf{m} \neq 0.$$

Because S is reverse well ordered, it has a largest element $\mathbf{m}_0$. Define

$$\text{mag}(T) = \mathbf{m}_0, \quad \text{dom}(T) = a_{\mathbf{m}_0} \mathbf{m}_0.$$

Then

$$T \sim \text{dom}(T).$$

The sign of T is the sign of $a_{\mathbf{m}_0}$.

9.2 Formal asymptotic relations

For nonzero transseries A, B define

$$\begin{aligned} A \prec B &\iff \text{mag}(A) \prec \text{mag}(B), \\ A \asymp B &\iff \text{mag}(A) = \text{mag}(B), \\ A \sim B &\iff A - B \prec B. \end{aligned}$$

The last condition means that A and B have the same dominant term, not merely the same dominant monomial.

Example 9.1. Let

$$A = 3x^2 + x, \quad B = 5x^2 - 7, \quad C = 3x^2 + e^{-x}.$$

Then $A \asymp B$ but $A \not\sim B$, since their leading coefficients differ. On the other hand, $A \sim C$: indeed, $A - C = x - e^{-x} \asymp x \prec x^2 \asymp A$. Thus A and C share the dominant term $3x^2$, even though their lower-order parts look very different.

9.3 Large, bounded, and infinitesimal elements

A transseries T is

- **large** if $|T| > r$ for every real r ;
- **bounded** if $|T| < r$ for some positive real r ;
- **infinitesimal** if $|T| < r$ for every positive real r .

In monomial language,

$$T \text{ large} \iff \text{mag}(T) \succ 1, \quad T \text{ infinitesimal} \iff \text{mag}(T) \prec 1.$$

A bounded transseries has the form

$$c + \varepsilon, \quad c \in \mathbb{R}, \quad \varepsilon \prec 1.$$

9.4 The natural valuation ring

The bounded elements form a valuation ring

$$\mathcal{O} = \{T : |T| \leq r \text{ for some } r \in \mathbb{R}_{>0}\}.$$

Its maximal ideal is the set of infinitesimals

$$\mathfrak{o} = \{T : T \prec 1\}.$$

The map

$$\text{st} : \mathcal{O} \rightarrow \mathbb{R}, \quad c + \varepsilon \mapsto c,$$

is the formal **standard part** or residue map.

Intuition

The valuation forgets coefficients and remembers order of magnitude. The residue map forgets infinitesimal corrections and remembers the constant part. Together, ordering and valuation turn asymptotic estimates into algebra.

9.5 Multiplicative normal form

Every nonzero T can be written uniquely as

$$T = a\mathfrak{m}(1 + \varepsilon), \quad a \in \mathbb{R}^\times, \quad \varepsilon \prec 1.$$

From this one obtains immediately:

$$\begin{aligned} T^{-1} &= a^{-1}\mathfrak{m}^{-1} \sum_{n \geq 0} (-\varepsilon)^n, \\ \log T &= \log a + \log \mathfrak{m} + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \varepsilon^n \quad (T > 0), \\ T^\alpha &= a^\alpha \mathfrak{m}^\alpha \sum_{n \geq 0} \binom{\alpha}{n} \varepsilon^n \quad (T > 0, \alpha \in \mathbb{R}). \end{aligned}$$

9.6 Leading-term arithmetic

If $A, B \neq 0$, then

$$\text{mag}(AB) = \text{mag}(A) \text{mag}(B), \quad \text{dom}(AB) = \text{dom}(A) \text{dom}(B).$$

For sums, the larger magnitude wins unless leading terms cancel. If $A \succ B$, then

$$A + B \sim A.$$

If $A \asymp B$, one must add the leading coefficients; cancellation may reveal a lower scale.

Worked example: cancellation exposes a hidden scale

Let

$$A = x + 1 + e^{-x}, \quad B = -x - 1 + x^{-3}.$$

Individually, $A \asymp B \asymp x$. But

$$A + B = x^{-3} + e^{-x} \sim x^{-3},$$

because $x^{-3} \succ e^{-x}$. Two cancellations move the answer from the scale x to the scale x^{-3} .

9.7 Formal limits

The order language lets us write analogues of limits:

$$T \rightarrow c \quad \text{means} \quad T - c \prec 1,$$

and

$$T \rightarrow +\infty \quad \text{means} \quad T > \mathbb{R}.$$

These are algebraic statements inside the transseries field. When a transseries represents a germ, they agree with the usual analytic interpretation.

Checkpoint

The entire formal calculus rests on two normal forms:

$$T = L + c + S \quad \text{and} \quad T = a\mathbf{m}(1 + \varepsilon).$$

The first is adapted to exponentiation; the second is adapted to inversion, logarithms, and powers.

Part III

Algebra, calculus, and composition

Formal arithmetic: familiar rules on an unfamiliar scale

10.1 Why the ordinary algebraic rules still work

A transseries is more elaborate than a polynomial, but its arithmetic is built from the same two operations:

$$\left(\sum_{\mathfrak{m}} a_{\mathfrak{m}} \mathfrak{m} \right) + \left(\sum_{\mathfrak{m}} b_{\mathfrak{m}} \mathfrak{m} \right) = \sum_{\mathfrak{m}} (a_{\mathfrak{m}} + b_{\mathfrak{m}}) \mathfrak{m},$$

and

$$\left(\sum_{\mathfrak{m}} a_{\mathfrak{m}} \mathfrak{m} \right) \left(\sum_{\mathfrak{n}} b_{\mathfrak{n}} \mathfrak{n} \right) = \sum_{\mathfrak{g}} \left(\sum_{\mathfrak{mn}=\mathfrak{g}} a_{\mathfrak{m}} b_{\mathfrak{n}} \right) \mathfrak{g}.$$

The support conditions from Chapter 6 are what make the inner coefficient sum finite. Once this finiteness theorem has been proved, addition and multiplication may be performed exactly as one would multiply ordinary formal power series.

Worked example: multiplying two sectors

Let

$$A = 1 + \frac{2}{x} + 3e^{-x}, \quad B = 1 - \frac{1}{x} + 5e^{-x}.$$

Then

$$AB = 1 + \frac{1}{x} - \frac{2}{x^2} + 8e^{-x} + \frac{7}{x}e^{-x} + 15e^{-2x}.$$

There is no mystery: collect equal monomials, but remember that $x^{-1}e^{-x}$ and e^{-x} are distinct monomials.

10.2 Inversion by separating the leading monomial

Suppose

$$T = a\mathfrak{m}(1 + \varepsilon), \quad \varepsilon \prec 1.$$

The formal geometric series gives

$$\frac{1}{T} = a^{-1}\mathfrak{m}^{-1} \left(1 - \varepsilon + \varepsilon^2 - \varepsilon^3 + \cdots \right).$$

This is legitimate because the family $(\varepsilon^n)_{n \geq 0}$ is summable: for every fixed monomial, only finitely many powers can contribute to its coefficient.

Worked example: an inverse containing two small scales

Write

$$T = x(1 + x^{-1} + xe^{-x}).$$

Set $u = x^{-1} + xe^{-x}$. Then

$$T^{-1} = x^{-1}(1 - u + u^2 - u^3 + \dots).$$

Keeping terms down to the scale x^{-3} and the first exponential sector gives

$$T^{-1} = x^{-1} - x^{-2} + x^{-3} - e^{-x} + 2x^{-1}e^{-x} - 3x^{-2}e^{-x} + xe^{-2x} + \dots.$$

The order in which one displays the terms is dictated by magnitude, not by the order in which the algebra produced them.

10.3 Infinite sums are allowed only when they are summable

The symbol

$$\sum_{i \in I} T_i$$

means a transseries only when the family is summable in the Hahn sense. Two conditions are useful to remember:

1. the union of all supports must be well based;
2. each fixed monomial must occur in only finitely many T_i .

For example,

$$\sum_{n \geq 0} x^{-n}$$

is summable, while

$$\sum_{n \geq 0} x^n$$

is not a legal expansion at $x = +\infty$. The second family keeps introducing larger monomials and has no largest term.

Common pitfall

The phrase “formal” does not mean “anything goes.” Formal transseries have strict support rules. Those rules replace numerical convergence and are what prevent ambiguous rearrangements.

10.4 A transseries parameter is a coefficient, not a monomial

Many differential equations produce a family

$$T(x; C) = \sum_{k \geq 0} C^k e^{-kAx} \Phi_k(x),$$

where C is an arbitrary constant. One should distinguish two viewpoints:

- for a fixed real or complex C , this is an ordinary transseries in x ;
- algebraically, one may work over the coefficient field $\mathbb{R}(C)$ or with a formal parameter C .

The exponential scale e^{-Ax} belongs to the monomial group. The integration constant C belongs to the coefficient field. Confusing these roles leads to errors when comparing magnitudes.

10.5 Truncation is an algebraic operation

If $\mathbf{n}$ is a monomial, write

$$T|_{\succeq \mathbf{n}}$$

for the sum of all terms of T whose monomials are at least $\mathbf{n}$. Because supports are well based, this is a legitimate transseries. Truncation is not generally a field homomorphism: multiplying two discarded tails can sometimes contribute at a retained scale. Nevertheless, if one keeps a sufficiently generous “guard band,” finite truncations can be computed reliably.

Core idea

Computer algebra with transseries is possible because every requested finite output depends on only finitely many input terms. The global object may be infinite, but a well-chosen cutoff turns each calculation into finite bookkeeping.

Exponentials, logarithms, and real powers

11.1 The three-part additive decomposition

Recall that every transseries has a unique decomposition

$$T = L + c + S,$$

where L is purely large, c is constant, and S is infinitesimal. Exponentiation separates these three scales:

$$e^T = \underbrace{e^L}_{\text{new monomial coefficient}} \underbrace{e^c}_{\text{unit}} \underbrace{\left(1 + S + \frac{S^2}{2!} + \cdots\right)}_{\text{unit}}.$$

This formula is the conceptual center of the LE construction.

Worked example: exponentiating a mixed transseries

Let

$$T = x^2 - 3x + 2 \log x + \log 7 + x^{-1} + e^{-x}.$$

Then

$$e^T = 7x^2 e^{x^2-3x} \exp(x^{-1} + e^{-x}).$$

To a few orders,

$$e^T = 7x^2 e^{x^2-3x} \left(1 + x^{-1} + \frac{x^{-2}}{2} + e^{-x} + x^{-1}e^{-x} + \frac{e^{-2x}}{2} + \cdots\right).$$

The very large part determines one monomial; only the small part is Taylor-expanded.

11.2 Why we do not Taylor-expand a large exponent

It would be meaningless at infinity to write

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots$$

as a transseries ordered from large to small: its support runs upward forever and has no dominant term. Instead, e^x is treated as a single monomial. Taylor series are used only for infinitesimal arguments.

This principle is worth memorizing:

large pieces create monomials; small pieces create Taylor expansions.

11.3 Logarithms by multiplicative normal form

For $T > 0$, write

$$T = a\mathfrak{m}(1 + \varepsilon), \quad a > 0, \quad \varepsilon \prec 1.$$

Then

$$\log T = \log a + \log \mathfrak{m} + \log(1 + \varepsilon),$$

where

$$\log(1 + \varepsilon) = \varepsilon - \frac{\varepsilon^2}{2} + \frac{\varepsilon^3}{3} - \cdots.$$

Worked example: taking a logarithm

Let

$$T = 3x^5 e^{\sqrt{x}} \left(1 + \frac{2}{x} - e^{-x}\right).$$

Set $u = 2x^{-1} - e^{-x}$. Then

$$\begin{aligned} \log T &= \log 3 + 5 \log x + \sqrt{x} + u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots \\ &= \sqrt{x} + 5 \log x + \log 3 + 2x^{-1} - 2x^{-2} + \frac{8}{3}x^{-3} \\ &\quad - e^{-x} + 2x^{-1}e^{-x} - 4x^{-2}e^{-x} - \frac{1}{2}e^{-2x} + \cdots. \end{aligned}$$

11.4 Real powers and binomial series

For $T > 0$ and $\alpha \in \mathbb{R}$, define

$$T^\alpha = \exp(\alpha \log T).$$

Equivalently,

$$(a\mathfrak{m}(1 + \varepsilon))^\alpha = a^\alpha \mathfrak{m}^\alpha \sum_{n \geq 0} \binom{\alpha}{n} \varepsilon^n.$$

This includes square roots, reciprocals, and arbitrary real powers.

Worked example: a square root across two scales

For

$$T = x^2 \left(1 + \frac{1}{x} + e^{-x}\right),$$

we obtain

$$\begin{aligned} \sqrt{T} &= x \left(1 + \frac{1}{2x} - \frac{1}{8x^2} + \frac{1}{16x^3} + \frac{1}{2}e^{-x} - \frac{1}{4x}e^{-x} \right. \\ &\quad \left. + \frac{3}{16x^2}e^{-x} - \frac{1}{8}e^{-2x} + \cdots\right). \end{aligned}$$

11.5 Identities and their domains

The familiar identities

$$\log(AB) = \log A + \log B, \quad e^{A+B} = e^A e^B$$

hold in the positive real transseries field. Complex logarithms require branch choices, exactly as in complex analysis. Even over the reals, $\log A$ is defined only for $A > 0$.

Common pitfall

The formal Taylor series for $\log(1 + S)$ requires $S \prec 1$, not merely $S < 1$ in some informal sense. If S contains a large monomial, factor the dominant monomial first.

Differentiation

12.1 Differentiate term by term

The derivative on ordinary monomials extends recursively:

$$(x^a)' = ax^{a-1}, \quad (e^L)' = L'e^L, \quad (\log x)' = x^{-1}.$$

For a transseries

$$T = \sum_{\mathfrak{m}} a_{\mathfrak{m}} \mathfrak{m},$$

define

$$T' = \sum_{\mathfrak{m}} a_{\mathfrak{m}} \mathfrak{m}'.$$

A nontrivial support theorem shows that this derivative family is summable. The resulting operation obeys the usual rules:

$$(A + B)' = A' + B', \quad (AB)' = A'B + AB', \quad (e^A)' = A'e^A, \quad (\log A)' = \frac{A'}{A}.$$

12.2 Logarithmic derivatives

For a nonzero monomial $\mathfrak{m}$, its logarithmic derivative is

$$\mathfrak{m}^\dagger := \frac{\mathfrak{m}'}{\mathfrak{m}} = (\log \mathfrak{m})'.$$

This often contains much less information than $\mathfrak{m}$ itself and is easier to compare. Examples are

$$\begin{aligned} (x^a)^\dagger &= \frac{a}{x}, \\ (e^{x^2})^\dagger &= 2x, \\ \left(x^a (\log x)^b e^{-x}\right)^\dagger &= -1 + \frac{a}{x} + \frac{b}{x \log x}. \end{aligned}$$

Thus

$$\left(x^a (\log x)^b e^{-x}\right)' = x^a (\log x)^b e^{-x} \left(-1 + \frac{a}{x} + \frac{b}{x \log x}\right).$$

12.3 Differentiation can change the comparison scale

If $A \prec B$, it is tempting to assume $A' \prec B'$. This is often true in tame settings but is not a purely formal consequence of $A \prec B$. For instance,

$$A = 1, \quad B = x$$

satisfies $A \prec B$, yet $A' = 0$ and $B' = 1$. A more dramatic issue in arbitrary function spaces is rapid oscillation, although the standard ordered transseries field excludes such behavior.

The H-field axiom gives the right replacement: if F is larger than every constant, then $F' > 0$. In particular, large positive transseries are eventually increasing in their germ interpretation.

12.4 Constants of differentiation

In the standard real transseries field,

$$T' = 0 \implies T \in \mathbb{R}.$$

This is not automatic for every differential extension, but it is a defining strength of the standard construction. Thus integration constants behave exactly as expected.

Worked example: differentiating a two-sector transseries

Let

$$T = x^3 + \log x + e^{-x}x^2 \left(1 - \frac{4}{x} - \frac{8}{x^3}\right).$$

For the exponential sector write $T_1 = e^{-x}x^2H$, where $H = 1 - 4x^{-1} - 8x^{-3}$. Then

$$T'_1 = e^{-x}x^2 \left(H' + \left(-1 + \frac{2}{x}\right)H \right).$$

Hence

$$T' = 3x^2 + x^{-1} + e^{-x}x^2 \left(-1 + \frac{6}{x} - \frac{4}{x^2} + \frac{8}{x^3} - \frac{8}{x^4} + \frac{24}{x^5} \right).$$

The logarithmic-derivative method avoids repeatedly applying the product rule.

12.5 A proof of the chain rule in the small-perturbation case

Let F be represented by a formal series around G and let $U \prec G$ be small enough that

$$F(G + U) = \sum_{n \geq 0} \frac{F^{(n)}(G)}{n!} U^n$$

is summable. Differentiating term by term gives

$$\begin{aligned} \frac{d}{dx} F(G + U) &= (G' + U') \sum_{n \geq 0} \frac{F^{(n+1)}(G)}{n!} U^n \\ &= (G' + U') F'(G + U). \end{aligned}$$

The general transseries chain rule follows by the recursive construction of composition.

Integration and asymptotic antidifferentiation

13.1 The simplest cases

For $a \neq -1$,

$$\int x^a dx = \frac{x^{a+1}}{a+1},$$

while

$$\int x^{-1} dx = \log x.$$

For a rapidly varying exponential, integration by parts naturally produces a transseries. If $L' \neq 0$, seek

$$\int e^L A dx = e^L B.$$

Differentiation yields the first-order equation

$$B' + L'B = A.$$

If $B'/L'B$ is small, solve recursively:

$$B = \frac{A}{L'} - \frac{1}{L'} \left(\frac{A}{L'} \right)' + \frac{1}{L'} \left[\frac{1}{L'} \left(\frac{A}{L'} \right)' \right]' - \cdots.$$

This is repeated integration by parts written as an operator expansion.

Worked example: the large- x expansion of an exponential integral

Take $L = x^2$ and $A = 1$. Then $L' = 2x$, so

$$\begin{aligned} \int^x e^{t^2} dt &\sim_{\text{as}} \frac{e^{x^2}}{2x} \left(1 + \frac{1}{2x^2} + \frac{3}{(2x^2)^2} + \frac{15}{(2x^2)^3} + \cdots \right) \\ &= \frac{e^{x^2}}{2x} \left(1 + \frac{1}{2x^2} + \frac{3}{4x^4} + \frac{15}{8x^6} + \cdots \right). \end{aligned}$$

A derivative check determines the recurrence. If $B = \sum_{n \geq 0} b_n x^{-2n-1}$, the equation $B' + 2xB = 1$ gives

$$b_0 = \frac{1}{2}, \quad b_{n+1} = \frac{2n+1}{2} b_n.$$

13.2 Asymptotic integration of a monomial

Let $\mathfrak{m} = e^L$. We want a transseries F with $F' = \mathfrak{m}$. The leading guess is

$$F_0 = \frac{\mathfrak{m}}{L'}.$$

Indeed,

$$F'_0 = \mathfrak{m} \left(1 - \frac{L''}{(L')^2} \right).$$

When $L''/(L')^2 \prec 1$, the relative error is small and may be corrected iteratively. This covers many familiar cases, such as $L = x^p$ with $p > 0$.

There is also a “slow” regime. For example, integrating x^a is not best viewed as dividing by $(a \log x)' = a/x$ when $a = -1$, because a resonance occurs and the answer moves to the new logarithmic scale $\log x$.

13.3 Exact closure is a deep theorem

The preceding algorithms produce formal asymptotic antiderivatives. A major theorem says more: the standard field of logarithmic-exponential transseries is closed under integration. Thus for every $A \in \mathbb{T}$ there is $B \in \mathbb{T}$ with $B' = A$.

This statement is not proved by naively integrating every term independently. Derivatives can mix neighboring monomials, and resonant logarithms may be required. The proof uses the structure of asymptotic couples and is one of the gateways from elementary transseries calculation to differential algebra [23, 2].

13.4 Formal definite integrals require endpoints

An antiderivative is a transseries near infinity. A numerical definite integral such as

$$\int_0^\infty f(x) \, dx$$

involves behavior at both 0 and ∞ and generally cannot be recovered from a transseries at infinity alone. Two functions can have the same complete asymptotic expansion at infinity but different integrals because they differ on a compact interval.

Common pitfall

A transseries is local asymptotic data. It may determine a germ near infinity without determining a global function or a global integral.

Composition and Taylor expansion

14.1 When composition makes asymptotic sense

If F is a transseries at $+\infty$, then $F \circ G$ should be defined when the inner transseries G is positive and infinitely large:

$$G > r \quad \text{for every } r \in \mathbb{R}.$$

This mirrors ordinary asymptotics: a large- x expansion for $F(x)$ can be substituted into only when $G(x) \rightarrow +\infty$.

For monomials, composition begins with

$$x^a \circ G = G^a, \quad e^L \circ G = \exp(L \circ G), \quad \log x \circ G = \log G.$$

It then extends to summable series.

14.2 Taylor's formula is the workhorse

Suppose

$$G = G_0 + U, \quad U \prec G_0,$$

and the derivatives of F form a summable Taylor family. Then

$$F(G_0 + U) = \sum_{n \geq 0} \frac{F^{(n)}(G_0)}{n!} U^n.$$

Unlike numerical Taylor series, this identity is controlled by support rather than by a radius of convergence.

Worked example: composing an exponential with a small correction

Let

$$F(x) = e^{-x}, \quad G = x + \log x + x^{-1}.$$

Then directly,

$$F \circ G = e^{-x} x^{-1} e^{-x^{-1}} = e^{-x} x^{-1} \left(1 - x^{-1} + \frac{x^{-2}}{2} - \frac{x^{-3}}{6} + \cdots \right).$$

The inner logarithm changes the algebraic prefactor; the infinitesimal x^{-1} is Taylor-expanded.

Worked example: Taylor composition around a large base

Let $F(x) = \log x$ and

$$G = x^2 + x + e^{-x} = x^2(1 + u), \quad u = x^{-1} + x^{-2}e^{-x}.$$

Then

$$\begin{aligned} \log G &= 2 \log x + \log(1 + u) \\ &= 2 \log x + x^{-1} - \frac{x^{-2}}{2} + \frac{x^{-3}}{3} + x^{-2}e^{-x} - x^{-3}e^{-x} + \dots \end{aligned}$$

14.3 Associativity and the chain rule

On its natural domain of positive infinite inner arguments, transseries composition satisfies

$$(F \circ G) \circ H = F \circ (G \circ H)$$

and

$$(F \circ G)' = (F' \circ G)G'.$$

The proof is not just a symbolic slogan: one must show that every family created by substitution is summable and that the result is independent of the chosen construction stage. Edgar's work gives a particularly approachable treatment in the well-based setting [18].

14.4 Composition is not substitution into an arbitrary small quantity

The expression $F(1/x)$ cannot in general be obtained from a transseries expansion of F at $+\infty$, because $1/x \rightarrow 0$. One needs a separate expansion of F near zero. Similarly, substituting a negative infinite transseries may leave the real logarithmic-exponential domain.

Checkpoint

For composition, always ask two questions:

1. Toward which endpoint is the outer expansion valid?
2. Does the inner transseries approach that endpoint in the required sector or direction?

Formal syntax does not erase analytic domain information.

Compositional inversion and implicit equations

15.1 The inverse problem

Given a positive infinite transseries F , we seek G such that

$$F \circ G = x, \quad G \circ F = x.$$

At a practical level, the process resembles series reversion:

1. invert the dominant behavior;
2. substitute the approximation;
3. measure the error;
4. add a smaller correction;
5. repeat.

15.2 Newton correction for composition

Suppose G is an approximate inverse and

$$E = F \circ G - x$$

is small compared with x . Replace G by $G + \Delta$. Taylor expansion gives

$$F(G + \Delta) - x = E + (F' \circ G)\Delta + \frac{1}{2}(F'' \circ G)\Delta^2 + \cdots.$$

The first Newton correction is therefore

$$\Delta = -\frac{E}{F' \circ G}.$$

Under the transseries ordering, the new error is usually quadratically smaller.

Worked example: inverting $x + \log x$

Let

$$F(x) = x + \log x.$$

The dominant inverse is $G_0 = x$. Since

$$F(G_0) - x = \log x,$$

we try $G_1 = x - \log x$. Then

$$\begin{aligned} F(G_1) - x &= -\log x + \log(x - \log x) \\ &= \log\left(1 - \frac{\log x}{x}\right) \\ &= -\frac{\log x}{x} - \frac{(\log x)^2}{2x^2} - \dots. \end{aligned}$$

The next correction begins with $(\log x)/x$. Continuing gives

$$F^{[-1]}(x) = x - \log x + \frac{\log x}{x} + \frac{(\log x)^2 - 2 \log x}{2x^2} + \dots.$$

Direct substitution verifies the displayed orders.

15.3 A structural theorem

Let

$$\mathcal{P} = \{F \in \mathbb{T} : F > \mathbb{R}\}$$

be the set of positive infinite transseries. With composition, $\mathcal{P}$ forms a group: every $F \in \mathcal{P}$ has a compositional inverse in $\mathcal{P}$. This is a deep closure property, not a consequence of ordinary field inversion.

15.4 Implicit equations by dominant balance

Consider

$$P(x, y) = 0$$

with transseries coefficients. A robust strategy is:

1. guess the dominant monomial of y by balancing the largest terms;
2. write $y = y_0(1 + u)$ or $y = y_0 + u$ with u smaller;
3. expand the equation in u ;
4. solve recursively for the next term.

Worked example: an algebraic equation with exponentially small roots

Solve

$$y^2 - y = e^{-x}.$$

The exact roots are

$$y_{\pm} = \frac{1 \pm \sqrt{1 + 4e^{-x}}}{2}.$$

Using the binomial series,

$$\begin{aligned} y_+ &= 1 + e^{-x} - e^{-2x} + 2e^{-3x} - 5e^{-4x} + 14e^{-5x} - \dots, \\ y_- &= -e^{-x} + e^{-2x} - 2e^{-3x} + 5e^{-4x} - 14e^{-5x} + \dots. \end{aligned}$$

The coefficients are Catalan numbers with alternating signs. The two branches arise from two different dominant balances, $y \sim 1$ and $y \sim 0$.

15.5 Implicit differentiation and transseries branches

If $P_y(x, y_0)$ is nonzero and dominates the nonlinear corrections, a transseries version of the implicit-function method constructs a branch near y_0 . The formal Newton step is

$$y_{n+1} = y_n - \frac{P(x, y_n)}{P_y(x, y_n)}.$$

Each step removes the current dominant residual. This “cancel the largest error” principle will recur in every differential-equation example.

How to compute a finite transseries in practice

16.1 The cancellation algorithm

Most hand calculations follow the same loop.

1. Choose a scale or ratio set, such as

$$\boldsymbol{\mu} = (x^{-1}, e^{-x}).$$

2. Guess the dominant term from the equation.
3. Substitute the partial transseries.
4. Find the largest uncanceled residual monomial.
5. Add one term whose image under the linearized operator cancels it.
6. Stop when the residual is below the requested cutoff.

This is long division, Newton's method, and asymptotic matching rolled into one.

16.2 A coefficient-matching template

Suppose an equation suggests

$$y = \sum_{n=0}^{\infty} a_n x^{-n}.$$

Substitute and rewrite every expression in the same monomial basis. If the result is

$$\sum_{n \geq 0} R_n(a_0, \dots, a_n) x^{-n} = 0,$$

then solve

$$R_n = 0$$

sequentially. Triangularity is the reason this works: the coefficient at a new scale usually depends linearly on the newest unknown and polynomially on earlier ones.

16.3 Several sectors

If the homogeneous linearization contains an exponentially small solution $e^{-Ax}x^\beta$, enlarge the ansatz to

$$y = \sum_{k \geq 0} C^k e^{-kAx} x^{k\beta} \sum_{n \geq 0} a_{k,n} x^{-n}.$$

The integer k labels the **nonperturbative sector**. Substitution often yields a two-dimensional triangular recurrence:

- solve sector $k = 0$ first;
- solve $k = 1$ using the linearized homogeneous equation;
- nonlinear products of lower sectors force $k = 2, 3, \dots$

16.4 Bookkeeping with weights

Assign a weight to each generator. For example,

$$w(x^{-1}) = 1, \quad w(e^{-x}) = M,$$

where M is chosen larger than any algebraic order being retained. Then sort monomials by the pair

$$(k, n) \text{extfore}^{-kx} x^{-n},$$

first by sector k , then by power n , or according to the exact asymptotic order needed. This is a practical computational device, not a replacement for the true monomial ordering.

16.5 A pseudocode skeleton

```
input: equation E[y] = 0, ansatz generators, cutoff
T := dominant approximation
repeat:
  R := expand E[T] down to cutoff
  if R has no retained term: return T
  r*m := dominant term of R
  L := linearization of E about T
  choose correction c*n so dominant(L[c*n]) = -r*m
  T := T + c*n
end
```

For nonlinear equations, recomputing the linearization after each correction is safest. For a fixed power-series ansatz, direct coefficient recurrences are faster.

16.6 Verification by residual

A finite transseries approximation should never be trusted merely because its coefficients look plausible. Substitute it back into the original equation and display the leading residual. If

$$E[T_N] = O(n),$$

then the approximation is certified through all monomials strictly larger than n , provided the linearization is nondegenerate on the omitted scale.

Checkpoint

A good transseries calculation has three visible parts:

1. the proposed support or sectors;
2. the recurrence or cancellation rule;
3. a residual check at the first omitted order.

The applications in the next part follow this pattern repeatedly.

Part IV

Worked examples and applications

A gallery of elementary transseries

The best way to make the notation natural is to see many short examples. In each case, the main task is to factor out the dominant scale and expand only an infinitesimal remainder.

17.1 Hyperbolic functions

Since

$$\sinh x = \frac{e^x}{2}(1 - e^{-2x}),$$

we have the exact transseries identity

$$\log(\sinh x) = x - \log 2 + \log(1 - e^{-2x}) = x - \log 2 - \sum_{n \geq 1} \frac{e^{-2nx}}{n}.$$

Thus

$$\log(\sinh x) = x - \log 2 - e^{-2x} - \frac{1}{2}e^{-4x} - \frac{1}{3}e^{-6x} - \dots.$$

Here the series converges for $x > 0$, but it is also a perfectly good transseries at infinity.

Similarly,

$$\begin{aligned} \tanh x &= \frac{1 - e^{-2x}}{1 + e^{-2x}} \\ &= 1 - 2e^{-2x} + 2e^{-4x} - 2e^{-6x} + \dots. \end{aligned}$$

17.2 A reciprocal with exponential nesting

Factor the dominant term:

$$\frac{1}{e^x + x} = \frac{e^{-x}}{1 + xe^{-x}}.$$

Because $xe^{-x} \prec 1$,

$$\frac{1}{e^x + x} = e^{-x} - xe^{-2x} + x^2e^{-3x} - x^3e^{-4x} + \dots.$$

The powers of x grow within successive exponential sectors, but the whole sequence still decreases because every additional factor e^{-x} overwhelms any fixed power of x .

17.3 Nested logarithms

For

$$F(x) = \log(x + \log x),$$

factor x :

$$\begin{aligned} F(x) &= \log x + \log \left(1 + \frac{\log x}{x} \right) \\ &= \log x + \frac{\log x}{x} - \frac{(\log x)^2}{2x^2} + \frac{(\log x)^3}{3x^3} - \dots \end{aligned}$$

This example already lies beyond ordinary Laurent series because the coefficients contain powers of $\log x$.

17.4 An exponential of an asymptotic series

Let

$$A = x + \frac{1}{x} + \frac{2}{x^2}.$$

Then

$$\begin{aligned} e^A &= e^x \exp \left(x^{-1} + 2x^{-2} \right) \\ &= e^x \left(1 + x^{-1} + \frac{5}{2}x^{-2} + \frac{13}{6}x^{-3} + \frac{73}{24}x^{-4} + \dots \right). \end{aligned}$$

A large exponential factor and a small power-series factor are kept conceptually separate.

17.5 A square root near a large branch

For fixed a ,

$$\begin{aligned} \sqrt{x^2 + a} &= x \sqrt{1 + ax^{-2}} \\ &= x + \frac{a}{2x} - \frac{a^2}{8x^3} + \frac{a^3}{16x^5} - \frac{5a^4}{128x^7} + \dots \end{aligned}$$

If instead the radicand contains a beyond-all-orders term,

$$\sqrt{x^2 + e^{-x}} = x + \frac{e^{-x}}{2x} - \frac{e^{-2x}}{8x^3} + \frac{e^{-3x}}{16x^5} - \dots$$

17.6 An inverse function close to a logarithm

Let $F(t) = e^t + t$ and write its inverse as $t = G(x)$. Since e^t dominates t , the first approximation is $G \sim \log x$. Put

$$G = \log x + u, \quad u \prec 1.$$

The equation $e^G + G = x$ becomes

$$xe^u + \log x + u = x,$$

or

$$e^u = 1 - \frac{\log x + u}{x}.$$

Taking logarithms and solving recursively gives

$$\begin{aligned} G(x) = \log x - \frac{\log x}{x} + \frac{\log x - \frac{1}{2}(\log x)^2}{x^2} \\ + \frac{-\log x + \frac{3}{2}(\log x)^2 - \frac{1}{3}(\log x)^3}{x^3} + \dots . \end{aligned}$$

A residual substitution is an efficient way to verify each displayed coefficient.

17.7 What these examples teach

All six computations use the same small number of moves:

1. identify and factor the dominant monomial;
2. turn the remaining factor into $1 + \varepsilon$ with $\varepsilon \prec 1$;
3. apply a familiar Taylor series formally;
4. reorder the answer by asymptotic magnitude;
5. check the first omitted term by substitution.

Lambert's W function, step by step

18.1 The inverse of $w \mapsto we^w$

Lambert's function W is defined by

$$W(x)e^{W(x)} = x.$$

For $x \rightarrow +\infty$, taking logarithms gives

$$W + \log W = \log x.$$

The first term must therefore be close to $\log x$, but substituting $W \sim \log x$ leaves an error $\log \log x$. This forces a second scale.

Set

$$L_1 = \log x, \quad L_2 = \log \log x.$$

The first two terms are

$$W \sim L_1 - L_2.$$

This is the simplest famous example in which iterated logarithms arise naturally from inversion.

18.2 Deriving the correction equation

Write

$$W = L_1 - L_2 + u, \quad u \prec L_2.$$

Then

$$\begin{aligned} \log W &= \log L_1 + \log \left(1 - \frac{L_2 - u}{L_1} \right) \\ &= L_2 + \log \left(1 - \frac{L_2 - u}{L_1} \right). \end{aligned}$$

The equation $W + \log W = L_1$ reduces exactly to

$$u + \log \left(1 - \frac{L_2 - u}{L_1} \right) = 0.$$

Now $L_2/L_1 \prec 1$, so the logarithm may be expanded.

18.3 Coefficient matching

Seek

$$u = \frac{P_1(L_2)}{L_1} + \frac{P_2(L_2)}{L_1^2} + \frac{P_3(L_2)}{L_1^3} + \dots,$$

where P_n is a polynomial. Substitution gives successively

$$\begin{aligned} P_1(z) &= z, \\ P_2(z) &= \frac{z^2}{2} - z, \\ P_3(z) &= \frac{z^3}{3} - \frac{3}{2}z^2 + z, \\ P_4(z) &= \frac{z^4}{4} - \frac{11}{6}z^3 + 3z^2 - z. \end{aligned}$$

Therefore

$$\begin{aligned} W(x) &= L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} \\ &\quad + \frac{L_2^3/3 - (3/2)L_2^2 + L_2}{L_1^3} \\ &\quad + \frac{L_2^4/4 - (11/6)L_2^3 + 3L_2^2 - L_2}{L_1^4} + \dots. \end{aligned} \tag{18.1}$$

Worked example: a numerical sanity check

At $x = 10^6$,

$$L_1 \approx 13.8155, \quad L_2 \approx 2.6258.$$

The successive approximations are roughly

$$11.1897, \quad 11.3798, \quad 11.3833, \quad 11.38336,$$

while the true principal value is about 11.38336. The point is not the decimal itself but the rapid gain obtained by respecting the correct logarithmic scale.

18.4 A compact all-orders formula

The polynomials can be expressed using unsigned Stirling numbers of the first kind. One standard form is

$$W = L_1 - L_2 + \sum_{\ell \geq 0} \sum_{m \geq 1} \frac{(-1)^\ell}{m!} \begin{bmatrix} \ell + m \\ \ell + 1 \end{bmatrix} \frac{L_2^m}{L_1^{\ell+m}}.$$

Here $\begin{bmatrix} n \\ k \end{bmatrix}$ counts permutations of n objects with k cycles. The double sum is ordered by total inverse-logarithmic degree. A classical analytic account of the function and its branches is [9].

18.5 The lower real branch

For $-1/e < x < 0$, the equation has two real branches. As $x \rightarrow 0^-$, the branch $W_{-1}(x)$ tends to $-\infty$. The same logarithmic mechanism applies, but branch choices must be tracked:

$$L_1 = \log(-x) < 0, \quad L_2 = \log(-L_1).$$

Then

$$W_{-1}(x) = L_1 - L_2 + \frac{L_2}{L_1} + \cdots.$$

The algebraic pattern is the same as Equation (18.1); the analytic branch is different.

18.6 Why this example matters

Lambert W displays three central features of transseries:

1. compositional inversion creates new scales;
2. logarithms iterate because the correction to a logarithm is another logarithm;
3. coefficient functions are polynomials in slower variables such as $\log \log x$.

It is a model for much more complicated inverse problems.

A linear differential equation with a hidden exponential

19.1 The equation

Consider

$$y'' + xy' + y = 0.$$

The left side factors as

$$\frac{d}{dx}(y' + xy),$$

so one integration gives

$$y' + xy = C_2.$$

Using the integrating factor $e^{x^2/2}$,

$$y(x) = C_1 e^{-x^2/2} + C_2 e^{-x^2/2} \int^x e^{t^2/2} dt.$$

19.2 The algebraic sector

Seek a formal inverse-power solution

$$y_0 = \sum_{n \geq 0} a_n x^{-2n-1}.$$

Substitution into $y' + xy = 1$ gives

$$a_0 = 1, \quad a_{n+1} = (2n+1)a_n.$$

Hence

$$y_0 = \frac{1}{x} + \frac{1}{x^3} + \frac{3}{x^5} + \frac{15}{x^7} + \frac{105}{x^9} + \cdots,$$

with

$$a_n = (2n-1)!! = \frac{(2n)!}{2^n n!}.$$

This series diverges factorially but is asymptotic to the second exact solution.

19.3 The invisible homogeneous solution

Every function

$$y = y_0 + Ce^{-x^2/2}$$

has the same inverse-power asymptotic expansion, because

$$e^{-x^2/2} \prec x^{-N} \quad \text{for every } N.$$

The full formal information is therefore the one-parameter transseries

$$y \sim \left(\frac{1}{x} + \frac{1}{x^3} + \frac{3}{x^5} + \cdots \right) + Ce^{-x^2/2}.$$

In this linear example there are no higher powers of $Ce^{-x^2/2}$ because superposition is linear.

Core idea

An ordinary asymptotic series describes a quotient space of solutions: it identifies functions that differ by something beyond all retained orders. A transseries restores the missing homogeneous coordinate C .

19.4 Divergence predicts the hidden scale

The n th algebraic term is approximately

$$\frac{(2n)!}{2^n n! x^{2n+1}}.$$

Stirling's formula shows that the smallest term occurs near $n \approx x^2/2$ and has size comparable, up to an algebraic factor, to $e^{-x^2/2}$. Thus the late terms of the algebraic sector know the exponential scale of the missing homogeneous solution.

19.5 Residual verification

Truncate after N terms:

$$y_N = \sum_{n=0}^{N-1} (2n-1)!! x^{-2n-1}.$$

A direct calculation gives

$$y'_N + xy_N = 1 - (2N-1)!! x^{-2N}.$$

Differentiating once more verifies Section 19.1 up to the corresponding first omitted order. This exact residual formula makes the asymptotic status transparent.

A nonlinear ODE and its tower of sectors

20.1 The model problem

Consider the Riccati-type equation

$$y' + y = \frac{1}{x} + y^2.$$

It is simple enough for hand computation but rich enough to generate a genuine nonlinear transseries.

20.2 The perturbative sector

Assume

$$y_0 = \sum_{m \geq 1} a_m x^{-m}.$$

Matching x^{-1} gives $a_1 = 1$. For $m \geq 2$,

$$a_m = (m-1)a_{m-1} + \sum_{j=1}^{m-1} a_j a_{m-j}.$$

The first coefficients are

$$a_1, a_2, a_3, a_4, a_5, a_6, a_7 = 1, 2, 8, 44, 296, 2312, 20384.$$

Thus

$$y_0 = \frac{1}{x} + \frac{2}{x^2} + \frac{8}{x^3} + \frac{44}{x^4} + \frac{296}{x^5} + \cdots.$$

The rapid coefficient growth warns us that this is an asymptotic, not convergent, series.

20.3 Linearization reveals the first exponential

Put

$$y = y_0 + \delta.$$

Using the equation for y_0 gives

$$\delta' + (1 - 2y_0)\delta = \delta^2.$$

At first order, discard δ^2 :

$$\delta'_1 + (1 - 2y_0)\delta_1 = 0.$$

Hence

$$\delta_1 = C \exp \left(-x + 2 \int y_0 dx \right).$$

Since $y_0 = x^{-1} + 2x^{-2} + \dots$,

$$\delta_1 = C e^{-x} x^2 h_1(x),$$

where

$$h_1 = 1 - \frac{4}{x} - \frac{8}{x^3} - \frac{52}{x^4} - \frac{384}{x^5} - \frac{3200}{x^6} - \dots.$$

The power x^2 comes from integrating the term $2/x$ in $2y_0$.

20.4 Nonlinearity forces higher sectors

Because the right side of Section 20.3 contains δ^2 , a first sector proportional to Ce^{-x} produces a second sector proportional to C^2e^{-2x} , which then interacts to produce all higher sectors. The natural ansatz is

$$y = y_0 + \sum_{k \geq 1} C^k e^{-kx} x^{2k} h_k(x), \quad h_k(x) = \sum_{n \geq 0} c_{k,n} x^{-n}.$$

For $k = 1$, the equation is homogeneous. For $k \geq 2$,

$$h'_k + \left(1 - k + \frac{2k}{x} - 2y_0 \right) h_k = \sum_{i=1}^{k-1} h_i h_{k-i}.$$

For example,

$$h_2 = -1 + \frac{6}{x} - \frac{6}{x^2} + \frac{8}{x^3} + \frac{48}{x^4} + \frac{384}{x^5} + \dots.$$

20.5 Where the free parameter enters

The perturbative recurrence Section 20.2 fixes every a_m . The first exponential equation is homogeneous, so its overall coefficient C is free. Every later sector is forced polynomially by the earlier ones. This is the typical one-parameter pattern for a first-order nonlinear ODE.

Worked example: checking the first sector

Let

$$q = e^{-x} x^2, \quad h_1 = 1 - 4x^{-1} + O(x^{-2}).$$

Then

$$\frac{q'}{q} = -1 + \frac{2}{x}.$$

Since $1 - 2y_0 = 1 - 2x^{-1} - 4x^{-2} + \dots$, the coefficient of qh_1 in the linearized left side is

$$\left(-1 + \frac{2}{x} \right) + (1 - 2y_0) = -4x^{-2} + \dots.$$

The derivative of h_1 begins with $4x^{-2}$, so the leading residual cancels. This explains the coefficient $-4/x$ without solving the entire recurrence.

20.6 Formal solution versus actual solution

The formal family Section 20.4 does not by itself say which value of C corresponds to a particular boundary condition, nor whether the sectors can be summed along a chosen complex direction. Those are analytic questions. Borel summation and Stokes phenomena, developed in Part V, explain how the formal parameter can jump when the direction of summation changes.

WKB analysis and the Airy equation

21.1 Why exponentials appear in second-order equations

Consider

$$\varepsilon^2 y''(x) = Q(x)y(x), \quad 0 < \varepsilon \ll 1.$$

If we try $y = \exp(S/\varepsilon)$, then

$$\frac{y''}{y} = \frac{(S')^2}{\varepsilon^2} + \frac{S''}{\varepsilon}.$$

Equation Section 21.1 becomes

$$(S')^2 + \varepsilon S'' = Q.$$

The leading balance is

$$S'_0 = \pm \sqrt{Q},$$

so one expects exponential scales

$$\exp\left(\pm \frac{1}{\varepsilon} \int^x \sqrt{Q(t)} dt\right).$$

This is the origin of the two WKB sectors.

21.2 Logarithmic derivative and the Riccati equation

It is often cleaner to set

$$u = \frac{y'}{y}.$$

Then

$$y''/y = u' + u^2.$$

For the Airy equation

$$y'' = xy,$$

we obtain

$$u' + u^2 = x.$$

The dominant balance $u^2 \sim x$ gives two branches

$$u \sim \pm x^{1/2}.$$

21.3 Solving the Riccati expansion

Write

$$u_{\pm} = \pm x^{1/2} + v.$$

Substitution into Section 21.2 and matching descending powers of $x^{3/2}$ yields

$$\begin{aligned} u_+ &= x^{1/2} - \frac{1}{4x} - \frac{5}{32x^{5/2}} - \frac{15}{64x^4} - \frac{1105}{2048x^{11/2}} + \cdots, \\ u_- &= -x^{1/2} - \frac{1}{4x} + \frac{5}{32x^{5/2}} - \frac{15}{64x^4} + \frac{1105}{2048x^{11/2}} + \cdots. \end{aligned}$$

The exponents advance by $3/2$ because differentiating $x^{1/2}$ lowers the exponent by one, while multiplication by the leading $x^{1/2}$ shifts corrections by one half.

21.4 Integrating back to the Airy solutions

Since $u = (\log y)'$, integrate the decaying branch:

$$\log y_- = -\frac{2}{3}x^{3/2} - \frac{1}{4}\log x - \frac{5}{48x^{3/2}} + \frac{5}{64x^3} + \cdots.$$

Exponentiating gives

$$y_-(x) = C_- x^{-1/4} e^{-2x^{3/2}/3} \left(1 - \frac{5}{48x^{3/2}} + \frac{385}{4608x^3} - \frac{85085}{663552x^{9/2}} + \cdots \right).$$

With the normalization $C_- = 1/(2\sqrt{\pi})$, this is the classical expansion of $\text{Ai}(x)$ on the positive real axis. The growing branch has the scale

$$x^{-1/4} e^{+2x^{3/2}/3}.$$

21.5 Turning points

A point where $Q(x) = 0$ is a **turning point**. The WKB monomials involve $Q^{-1/4}$ and therefore become singular there. Near a simple turning point, rescaling reduces the differential equation to the Airy equation. Thus Airy transseries are not an isolated curiosity; they are the universal local model for a wide class of turning-point problems.

21.6 Dominance depends on complex direction

For complex $x = re^{i\theta}$,

$$x^{3/2} = r^{3/2} e^{3i\theta/2}.$$

The sign of

$$\Re\left(\frac{2}{3}x^{3/2}\right)$$

determines which WKB exponential grows and which decays. Along rays where this real part vanishes, the two exponentials have equal magnitude. These rays are tied to Stokes and anti-Stokes geometry. The formal sectors stay the same, but their analytic importance changes with direction. A classical rigorous treatment of these asymptotic constructions is given by Olver [27].

Core idea

WKB analysis can be summarized as follows: turn a differential equation into an equation for a logarithmic derivative, find the dominant algebraic branches, integrate them to create exponential monomials, and then solve recursively for the prefactor series.

21.7 Newton polygons in one paragraph

For more complicated differential equations, one may plot how candidate terms scale under an ansatz $y \asymp e^S x^\alpha$. Balancing terms that lie on a common edge of the resulting Newton polygon identifies possible dominant exponents. The polygon does not compute the full transseries, but it organizes which balances should be tried and which exponential scales can occur.

Singular perturbations and boundary layers

22.1 Regular versus singular perturbation

A parameter ε is **regular** if setting $\varepsilon = 0$ leaves an equation of the same essential type. It is **singular** if the highest derivative disappears, the order drops, or boundary conditions can no longer all be imposed. Singular perturbations routinely produce scales such as

$$e^{-A/\varepsilon},$$

which are invisible to every power of ε .

22.2 A first-order boundary layer revisited

The equation

$$\varepsilon y' + y = 1, \quad y(0) = 0,$$

has solution

$$y = 1 - e^{-x/\varepsilon}.$$

The outer expansion at fixed $x > 0$ is simply $y \sim 1$. To resolve the initial condition, introduce the inner variable

$$X = \frac{x}{\varepsilon}.$$

Then the equation becomes

$$\frac{dy}{dX} + y = 1,$$

whose solution $1 - e^{-X}$ is not small when $X = O(1)$. The same exponential is therefore negligible in the outer region and essential in the boundary layer.

22.3 A two-point example

Consider

$$\varepsilon y'' + y' = 0, \quad y(0) = 0, \quad y(1) = 1.$$

The exact solution is

$$y(x) = \frac{1 - e^{-x/\varepsilon}}{1 - e^{-1/\varepsilon}}.$$

Expanding the denominator geometrically gives the transseries

$$\begin{aligned} y(x) &= (1 - e^{-x/\varepsilon}) \left(1 + e^{-1/\varepsilon} + e^{-2/\varepsilon} + \dots \right) \\ &= 1 - e^{-x/\varepsilon} + e^{-1/\varepsilon} - e^{-(x+1)/\varepsilon} + e^{-2/\varepsilon} - \dots \end{aligned}$$

Several exponential actions occur: x , 1 , $x+1$, and so on. Which term is largest depends on the location x .

22.4 Outer, inner, and composite descriptions

A standard matched-asymptotic calculation has three levels:

1. the **outer expansion**, valid away from the thin layer;
2. the **inner expansion**, written in a stretched variable such as $X = x/\varepsilon$;
3. a **composite approximation**, obtained by adding outer and inner approximations and subtracting their common overlap.

Transseries offer a fourth viewpoint: retain both the algebraic and exponential scales in one symbolic object. This does not eliminate the need to state regions of validity, but it makes the hidden scales explicit.

22.5 Multiple boundary layers

If an equation has layers near several endpoints, one may encounter

$$e^{-x/\varepsilon}, \quad e^{-(1-x)/\varepsilon}, \quad e^{-1/\varepsilon},$$

and products of these monomials. The product

$$e^{-x/\varepsilon} e^{-(1-x)/\varepsilon} = e^{-1/\varepsilon}$$

shows that local layer sectors can interact to create a global exponentially small scale.

22.6 Nonuniformity is not a contradiction

For fixed $x > 0$,

$$e^{-x/\varepsilon} \prec \varepsilon^N \quad \text{for all } N,$$

but this estimate is not uniform down to $x = 0$. When $x = \varepsilon X$, the term becomes e^{-X} , which is order one. An asymptotic statement is incomplete unless its limiting process and domain are specified.

Common pitfall

Never say merely “ $e^{-x/\varepsilon}$ is negligible.” Say where: it is beyond all algebraic orders for fixed $x > 0$ as $\varepsilon \rightarrow 0^+$, but not inside the $x = O(\varepsilon)$ boundary layer.

Difference equations and Stirling's series

23.1 The gamma function as a difference equation

The defining recurrence

$$\Gamma(x+1) = x\Gamma(x)$$

becomes, after taking logarithms,

$$F(x+1) - F(x) = \log x, \quad F = \log \Gamma.$$

This is the difference-equation analogue of antidifferentiation.

23.2 The shift operator

Let $D = d/dx$. Formally,

$$F(x+1) = e^D F(x),$$

so Section 23.1 is

$$(e^D - 1)F = \log x.$$

Multiplying by $D/(e^D - 1)$ gives an equation for F' :

$$F' = \frac{D}{e^D - 1} \log x.$$

The generating function for Bernoulli numbers is

$$\frac{z}{e^z - 1} = \sum_{n \geq 0} B_n \frac{z^n}{n!}.$$

Therefore

$$F' = \sum_{n \geq 0} \frac{B_n}{n!} D^n \log x.$$

23.3 Integrating the formal expansion

Using $B_1 = -1/2$ and $B_{2n+1} = 0$ for $n \geq 1$, one obtains

$$F'(x) \sim_{\text{as}} \log x - \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} - \frac{1}{252x^6} + \cdots.$$

Integration yields

$$\log \Gamma(x) \sim_{\text{as}} \left(x - \frac{1}{2}\right) \log x - x + C + \sum_{n \geq 1} \frac{B_{2n}}{2n(2n-1)x^{2n-1}}. \quad (23.1)$$

The difference equation does not determine the constant C . The standard normalization of Γ gives

$$C = \frac{1}{2} \log(2\pi).$$

23.4 Why Stirling's series diverges

The Bernoulli numbers satisfy

$$|B_{2n}| \sim \frac{2(2n)!}{(2\pi)^{2n}}.$$

Consequently, the terms of Equation (23.1) eventually grow for every fixed x . The least term occurs at roughly

$$n \approx \pi x,$$

and its scale is approximately

$$e^{-2\pi x}$$

times an algebraic factor. Once again, factorial divergence predicts a beyond-all-orders exponential.

23.5 The periodic ambiguity

If F solves Section 23.1, then so does

$$F + P$$

for every period-one function $P(x+1) = P(x)$. In the ordered real LE-transseries field, nonconstant oscillatory periodic functions are absent, so only a constant ambiguity remains. In broader complex frameworks, Fourier modes and exponentially small Stokes data encode related phenomena.

23.6 Euler–Maclaurin as a transseries machine

The same operator identity underlies the Euler–Maclaurin formula. For a smooth f ,

$$\begin{aligned} \sum_{k=a}^b f(k) &= \int_a^b f(t) dt + \frac{f(a) + f(b)}{2} \\ &\quad + \sum_{n=1}^{N-1} \frac{B_{2n}}{(2n)!} \left(f^{(2n-1)}(b) - f^{(2n-1)}(a) \right) + R_N. \end{aligned}$$

When the derivative series diverges, optimal truncation and exponentially small corrections enter. Thus transseries are as natural for difference equations and sums as they are for differential equations.

Fractional iteration and Abel's equation

24.1 From integer iterates to a continuous parameter

For a function or transseries T , write

$$T^{[0]}(x) = x, \quad T^{[1]} = T, \quad T^{[n+m]} = T^{[n]} \circ T^{[m]}.$$

A **fractional iterate** is a family $T^{[s]}$ with real or complex s satisfying

$$T^{[s+t]} = T^{[s]} \circ T^{[t]}.$$

The central device is to convert composition into addition.

24.2 Abel's equation

Seek an **Abel coordinate** A satisfying

$$A(T(x)) = A(x) + 1.$$

Then define

$$T^{[s]}(x) = A^{[-1]}(A(x) + s).$$

Indeed,

$$\begin{aligned} T^{[s]}(T^{[t]}(x)) &= A^{[-1]}(A(T^{[t]}(x)) + s) \\ &= A^{[-1]}(A(x) + t + s) = T^{[s+t]}(x). \end{aligned}$$

24.3 The trivial example

For $T(x) = x + 1$, take $A(x) = x$. Then

$$T^{[s]}(x) = x + s.$$

The importance of Abel's equation is that the same idea works when no elementary coordinate is available.

24.4 A nontrivial example: $T(x) = x + 1/x$

The map

$$T(x) = x + \frac{1}{x}$$

increases x only slightly. Since

$$T(x)^2 - x^2 = 2 + x^{-2},$$

the leading Abel coordinate should be $x^2/2$. A correction by $\log x$ removes the next error. Solving Section 24.2 term by term gives

$$A(x) = \frac{x^2}{2} - \frac{1}{2} \log x + \frac{1}{8x^2} + \frac{5}{96x^4} + \frac{7}{288x^6} - \frac{1}{1280x^8} + \cdots. \quad (24.1)$$

Substitution of $x + x^{-1}$ into this expression verifies

$$A(T(x)) - A(x) = 1$$

to the displayed order.

24.5 The fractional iterate

Using Section 24.2 and reverting the series gives

$$\begin{aligned} T^{[s]}(x) = x + \frac{s}{x} - \frac{s(s-1)}{2x^3} + \frac{s(s-1)^2}{2x^5} \\ - \frac{s(s-1)(3s-2)(5s-7)}{24x^7} + O(x^{-9}). \end{aligned} \quad (24.2)$$

Several checks are immediate:

$$T^{[0]}(x) = x, \quad T^{[1]}(x) = x + x^{-1}.$$

For $s = 2$, Equation (24.2) gives

$$x + \frac{2}{x} - \frac{1}{x^3} + \frac{1}{x^5} - \frac{1}{x^7} + \cdots,$$

which agrees with directly expanding $T(T(x))$.

24.6 The iterative logarithm

Differentiating $T^{[s]}$ with respect to s at $s = 0$ defines the **iterative logarithm** or infinitesimal generator

$$V(x) = \left. \frac{\partial}{\partial s} T^{[s]}(x) \right|_{s=0}.$$

The flow equation is

$$\frac{\partial}{\partial s} T^{[s]}(x) = V(T^{[s]}(x)).$$

From Equation (24.2),

$$V(x) = \frac{1}{x} + \frac{1}{2x^3} + \frac{1}{2x^5} + \frac{7}{12x^7} + \cdots.$$

This is the composition-theoretic analogue of the logarithm of a matrix.

24.7 Existence and uniqueness subtleties

Abel coordinates are not automatically unique as functions: if P is period one, then $A + P \circ A$ may produce another coordinate under suitable conditions. In formal transseries classes, normalization and support restrictions often restore uniqueness. Edgar proved substantial existence results for large positive transseries of exponentiality zero and developed their fractional iterates within the transseries framework [19].

Optional advanced viewpoint

Iteration near a fixed point is usually linearized by Schröder's equation rather than Abel's equation. Iteration near infinity, where $T(x) - x$ is small or of controlled growth, naturally leads to Abel coordinates. Tetration and continuous iteration of the exponential require still richer asymptotic structures and delicate choices of domain.

Part V

From formal expressions to analytic functions

Formal transseries versus analytic functions

25.1 A formal object is not automatically a function

The expression

$$\tilde{F}(x) = \sum_{n \geq 0} n! x^{-n-1}$$

is a valid formal power-series sector, but it diverges for every finite x . It therefore cannot be interpreted by ordinary pointwise summation. A formal transseries records ordered asymptotic data; an analytic function is an object with numerical values on a domain. Connecting the two requires an additional theorem or summation procedure.

25.2 Asymptotic representation

Let

$$\tilde{F} = \sum_{j \geq 0} a_j \mathfrak{m}_j, \quad \mathfrak{m}_0 \succ \mathfrak{m}_1 \succ \cdots$$

An analytic function F is represented by this transseries on a domain if, for each retained cutoff,

$$F - \sum_{j < N} a_j \mathfrak{m}_j = o(\mathfrak{m}_{N-1})$$

uniformly in the specified limit or sector. For multiple exponential sectors, one must state how the transseries parameter is chosen and in which complex direction the comparison holds.

25.3 The asymptotic map is not injective

At infinity,

$$0 \quad \text{and} \quad e^{-x}$$

have the same expansion in inverse powers of x . More generally, adding any flat function smaller than all members of the selected scale leaves that restricted asymptotic expansion unchanged. Consequently, a Poincaré series alone usually cannot identify a unique function.

A fuller transseries distinguishes many flat corrections, but even a formal transseries may admit several analytic sums in different directions. Stokes phenomena measure this remaining ambiguity.

25.4 Sectors in the complex plane

For $x \in \mathbb{C}$, a monomial such as e^{-Ax} is small only where

$$\Re(Ax) > 0.$$

Thus analytic realization is usually sectorial. A series may be summable in one angular sector, fail to be summable on a boundary ray, and acquire a different exponentially small contribution after crossing that ray.

25.5 Three levels of assertion

It is helpful to distinguish the following statements:

1. *Formal*: substitution into an equation cancels every monomial.
2. *Asymptotic*: finite truncations approximate an actual function with controlled remainders.
3. *Summability*: a specified operation reconstructs a canonical analytic function from the formal series in a chosen direction.

A formal calculation can be correct even when summability has not been proved. Conversely, numerical agreement does not replace a formal residual calculation.

25.6 Why summation should respect algebra and calculus

An ideal summation map $\mathcal{S}$ should satisfy, whenever both sides make sense,

$$\begin{aligned}\mathcal{S}(A + B) &= \mathcal{S}(A) + \mathcal{S}(B), \\ \mathcal{S}(AB) &= \mathcal{S}(A)\mathcal{S}(B), \\ \mathcal{S}(A') &= (\mathcal{S}A)', \\ \mathcal{S}(F \circ G) &= (\mathcal{S}F) \circ (\mathcal{S}G).\end{aligned}$$

No single elementary method covers every transseries, but Borel–Laplace summation achieves much of this program for Gevrey and resurgent classes.

Borel transformation and Laplace summation

26.1 Why factorial divergence suggests dividing by $n!$

Consider a formal inverse-power series

$$\tilde{\phi}(x) = \sum_{n=0}^{\infty} a_n x^{-n-1}.$$

If a_n grows like $n!$, the ordinary series diverges. The **formal Borel transform** removes this factorial growth:

$$\mathcal{B}\tilde{\phi}(p) = \hat{\phi}(p) := \sum_{n=0}^{\infty} \frac{a_n}{n!} p^n.$$

The new series often has a nonzero radius of convergence.

The inverse operation is suggested by

$$\int_0^{\infty} e^{-xp} p^n dp = \frac{n!}{x^{n+1}}, \quad \Re x > 0.$$

Thus the directional Laplace transform

$$\mathcal{L}_{\theta}\hat{\phi}(x) = \int_0^{e^{i\theta}\infty} e^{-xp} \hat{\phi}(p) dp$$

formally reconstructs the original powers.

26.2 The Borel–Laplace recipe

To sum $\tilde{\phi}$ in direction θ :

1. form the local Borel series Section 26.1;
2. analytically continue it along the ray $p \in e^{i\theta}\mathbb{R}_{\geq 0}$;
3. check that its growth is at most exponential along the ray;
4. apply the Laplace integral Section 26.1.

When all steps work, define

$$\mathcal{S}_{\theta}\tilde{\phi} := \mathcal{L}_{\theta}\mathcal{B}\tilde{\phi}.$$

26.3 Euler's alternating factorial series

Take

$$\tilde{F}_-(x) = \sum_{n \geq 0} (-1)^n n! x^{-n-1}.$$

Its Borel transform is the convergent geometric series

$$\hat{F}_-(p) = \sum_{n \geq 0} (-p)^n = \frac{1}{1+p}.$$

There is no singularity on the positive p -axis, so

$$\mathcal{S}_0 \tilde{F}_-(x) = \int_0^\infty \frac{e^{-xp}}{1+p} dp = -e^x \operatorname{Ei}(-x), \quad x > 0.$$

The Borel sum is exactly the function introduced in Chapter 25's motivating examples.

26.4 Watson's lemma: why the Laplace integral has the right expansion

Suppose $\hat{\phi}$ is analytic near $p = 0$ with

$$\hat{\phi}(p) = \sum_{n=0}^{N-1} \frac{a_n}{n!} p^n + R_N(p), \quad R_N(p) = O(p^N).$$

Insert this into the Laplace integral. The polynomial part gives

$$\sum_{n=0}^{N-1} a_n x^{-n-1}.$$

On a short interval, the remainder contributes $O(x^{-N-1})$ after rescaling $p = u/x$. On the remaining interval, the exponential kernel makes the contribution smaller still, under an exponential-growth bound. Therefore

$$\mathcal{L}_0 \hat{\phi}(x) \sim_{\text{as}} \sum_{n \geq 0} a_n x^{-n-1}.$$

This elementary argument is the core of Watson's lemma.

26.5 Gevrey order

A formal series is **Gevrey-1** if, roughly,

$$|a_n| \leq C A^n n!$$

for some $C, A > 0$. Then its Borel transform converges near the origin. More generally, growth like $\Gamma(1 + sn)$ leads to Gevrey order s and may require generalized Borel transforms. Most elementary resurgent examples are Gevrey-1 after a suitable change of variable.

26.6 Products become convolution

If

$$\tilde{f} = \sum a_n x^{-n-1}, \quad \tilde{g} = \sum b_n x^{-n-1},$$

then, up to conventions concerning constant terms,

$$\mathcal{B}(\tilde{f}\tilde{g}) = \hat{f} * \hat{g},$$

where

$$(\hat{f} * \hat{g})(p) = \int_0^p \hat{f}(q) \hat{g}(p-q) dq.$$

The Laplace transform turns convolution back into multiplication. This is why Borel–Laplace summation can respect nonlinear equations rather than merely sum isolated series.

26.7 Derivatives and multiplication by x

With the convention Section 26.1, differentiation and multiplication correspond to elementary Borel-plane operations. For example,

$$\mathcal{B}(\tilde{\phi}') = -p\hat{\phi}(p),$$

provided no polynomial or constant boundary term is present. Thus differential equations become integral or convolution equations in the Borel plane, where analytic continuation can often be studied directly.

Stokes phenomena: exponentially small jumps

27.1 A Borel singularity on the integration ray

Now consider the nonalternating factorial series

$$\tilde{F}_+(x) = \sum_{n \geq 0} n! x^{-n-1}.$$

Its Borel transform is

$$\hat{F}_+(p) = \frac{1}{1-p},$$

which has a pole at $p = 1$ on the positive integration ray. The ordinary Laplace integral is ambiguous. We define two **lateral sums**, one just above and one just below the ray:

$$\mathcal{S}_{0\pm} \tilde{F}_+ = \int_0^{\infty e^{\pm i0}} \frac{e^{-xp}}{1-p} dp.$$

27.2 Computing the jump

The difference of the two contours is a small loop around $p = 1$. With the orientation convention above,

$$\mathcal{S}_{0+} \tilde{F}_+ - \mathcal{S}_{0-} \tilde{F}_+ = 2\pi i e^{-x}.$$

The sign reverses if the lateral-orientation convention is reversed, but the scale and magnitude are fixed.

This formula is the simplest complete Stokes phenomenon:

- the two sums have the same power-series expansion;
- they differ by an exponential smaller than every power;
- the exponential scale is determined by the Borel singularity location $p = 1$.

27.3 Why the singularity location becomes an action

A singularity at $p = A$ contributes, after Laplace transformation, a factor

$$e^{-Ax}.$$

Indeed, shifting $p = A + q$ gives

$$e^{-xp} = e^{-Ax} e^{-xq}.$$

The number A is often called an **action**, especially in WKB theory and physics. A constellation of Borel singularities therefore predicts a constellation of exponential transmonomials.

27.4 The transseries parameter absorbs the jump

Suppose a formal solution has the form

$$\tilde{y} = \tilde{\Phi}_0 + C e^{-Ax} \tilde{\Phi}_1 + \dots.$$

If the lateral sum of $\tilde{\Phi}_0$ jumps by

$$S e^{-Ax} \mathcal{S} \tilde{\Phi}_1,$$

then the represented analytic solution remains unchanged when C jumps oppositely:

$$C_+ = C_- - S.$$

The constant S is a **Stokes constant**. The formal parameter is not a flaw; it is the coordinate that makes sectorial analytic continuation possible.

27.5 Stokes lines and anti-Stokes lines

Terminology varies between communities. A useful geometric distinction is:

- a ray where a Borel singularity obstructs directional summation is a Stokes direction in the Borel–Laplace sense;
- a ray where two exponentials have equal magnitude is often called an anti-Stokes line, although some authors exchange the names.

Always check the convention in a source. The invariant facts are which exponential is dominant, where the Laplace contour meets singularities, and what jump occurs.

27.6 A jump can be invisible to all perturbative orders

If

$$F_+ - F_- = 2\pi i e^{-x},$$

then for every N ,

$$F_+ - F_- = o(x^{-N}).$$

Thus no finite or infinite inverse-power expansion can detect the jump. The transseries sector e^{-x} is exactly the missing resolution.

27.7 Berry smoothing

An idealized Stokes jump looks discontinuous when one compares asymptotic expansions on the two sides of a ray. The underlying analytic function is usually smooth. Near optimal truncation, the effective multiplier of the small exponential often changes rapidly but smoothly according to an error-function profile. This phenomenon, discovered and developed by Berry and others, explains how a discontinuous asymptotic bookkeeping rule emerges from a smooth function.

Resurgence and alien calculus, gently

28.1 The central resurgent idea

A divergent series may become an analytic germ after Borel transformation. Continue that germ through the Borel plane. If its singularities are isolated and their local expansions are themselves related to other sectors of the original problem, the series is called **resurgent** in an appropriate technical sense.

The slogan is:

The different exponential sectors are not independent; each sector reappears in the singularity structure of the others.

28.2 A toy large-order relation

Let

$$\widehat{\phi}(p) = \frac{1}{1 - p/A}.$$

Expanding at $p = 0$ gives

$$\widehat{\phi}(p) = \sum_{n \geq 0} \frac{p^n}{A^n},$$

so the original formal coefficients are

$$a_n = \frac{n!}{A^n}.$$

The Borel singularity at A and the factorial-over-power growth of a_n are two views of the same fact. More complicated singularities produce refinements such as

$$a_n \sim \frac{S}{2\pi i} \frac{\Gamma(n + \beta)}{A^{n+\beta}} \left(b_0 + \frac{Ab_1}{n + \beta - 1} + \cdots \right).$$

The coefficients b_j typically belong to the neighboring exponential sector.

28.3 Reading another sector from late coefficients

Suppose the perturbative coefficients a_n are known to high order. One can:

1. estimate A from ratios such as na_{n-1}/a_n ;

2. remove the leading factorial-over-power growth;
3. extrapolate the remaining sequence to estimate Sb_0 ;
4. examine $1/n$ corrections to recover $b_1, b_2, \dots$

This is one reason large-order numerical data can reveal nonperturbative physics.

28.4 Alien derivatives as singularity detectors

Ordinary differentiation measures change with respect to x . An **alien derivative** Δ_A is designed to measure the singular behavior at Borel location A . Informally,

$$\Delta_A(\text{sector at action } 0) \propto \text{sector at action } A.$$

Alien derivatives satisfy algebraic rules and can be assembled into the Stokes automorphism. Their exact normalization depends on conventions and on whether one uses pointed alien derivatives, logarithms of Stokes maps, or related variants.

Optional advanced viewpoint

In a one-parameter problem, a bridge equation often has the schematic form

$$\dot{\Delta}_A \tilde{y}(x, C) = S_A(C) \frac{\partial}{\partial C} \tilde{y}(x, C).$$

The left side acts in the Borel-singularity direction; the right side acts on the transseries parameter. The equation “bridges” alien calculus and ordinary parameter calculus. Its precise polynomial $S_A(C)$ is problem dependent.

28.5 Multi-instanton and logarithmic sectors

A simple nonlinear problem may have sectors

$$e^{-kAx} \Phi_k(x), \quad k = 0, 1, 2, \dots$$

Resonances can force additional powers of $\log x$:

$$e^{-kAx} x^\beta (\log x)^j \Phi_{k,j}(x).$$

Several independent actions $A_1, \dots, A_r$ lead to a lattice

$$\exp(-(k_1 A_1 + \dots + k_r A_r)x).$$

This resembles a grid-based transseries, but analytic resurgence imposes extra relations between the grid sectors.

28.6 Not every transseries is resurgent

The formal LE field is enormous. Resurgent transseries form special analytic subclasses with controlled Borel continuation and singularities. Therefore:

- “transseries” does not automatically mean “resurgent”;
- an LE-transseries may be perfectly meaningful algebraically without a chosen Borel sum;
- a resurgent expansion may use complex actions and logarithmic sectors not most naturally viewed inside the simplest ordered real LE field.

The two theories overlap powerfully but should not be identified.

Optimal truncation, hyperasymptotics, and exponential improvement

29.1 Why more terms eventually hurt

Suppose

$$a_n \sim CA^{-n}\Gamma(n + \beta).$$

The magnitude of the n th term $a_n x^{-n}$ has ratio approximately

$$\frac{|a_{n+1}|x^{-n-1}}{|a_n|x^{-n}} \sim \frac{n + \beta}{|A|x}.$$

Terms decrease until $n \approx |A|x$ and grow afterward. Truncating near the least term is **optimal truncation**.

29.2 Estimating the least term

Stirling's formula gives, at $n \approx Ax$ with $A > 0$,

$$\Gamma(n + \beta)A^{-n}x^{-n} \asymp x^{\beta-1/2}e^{-Ax}$$

times a constant factor. Thus the optimally truncated error has exactly the scale predicted by the nearest Borel singularity.

29.3 First-level hyperasymptotics

Ordinary asymptotics expands a function around the Borel origin. Hyperasymptotics proceeds further:

1. truncate the first series near its least term;
2. express the remainder as a Laplace integral localized near the nearest Borel singularity;
3. expand the singularity contribution in its own asymptotic series;
4. truncate that new series optimally;

5. continue recursively to more distant singularities.

Each level resolves a smaller exponential scale.

29.4 Stirling's series as an example

For Equation (23.1), Bernoulli growth places the nearest relevant Borel singularities at distances related to 2π . Truncation near $n \approx \pi x$ leaves an error on the scale $e^{-2\pi x}$. Exponentially improved versions of Stirling's formula express the remainder in terms of terminant functions whose switching captures the Stokes phenomenon.

29.5 Supersymptotics versus hyperasymptotics

Terminology is useful:

- **supersymptotics** means optimally truncating the first divergent series;
- **hyperasymptotics** means re-expanding and optimally truncating the exponentially small remainder, possibly repeatedly.

A full transseries organizes all these levels in advance. Hyperasymptotics is the numerical strategy for extracting them efficiently; the classical sources include Dingle and Berry–Howls [14, 4].

29.6 Practical coefficient acceleration

When coefficients are known numerically, Richardson extrapolation, Borel–Padé approximation, conformal maps, and sequence transformations can estimate actions and Stokes constants. These are powerful tools, but they do not replace structural analysis: spurious Padé poles and finite-data effects can imitate singularities.

Checkpoint

The analytic story now fits into one chain:

$$\text{factorial divergence} \iff \text{Borel singularity} \iff \text{exponential scale} \iff \text{Stokes jump}.$$

Resurgence enriches this chain by identifying the local data at one singularity with the coefficients of another transseries sector.

Saddle points and transseries of integrals

30.1 Laplace-type integrals

Consider

$$I(x) = \int_{\Gamma} e^{-xV(t)} a(t) dt, \quad x \rightarrow +\infty.$$

The dominant contributions come from points where $\Re V$ is smallest along the contour: endpoints, poles, or critical points satisfying

$$V'(t_*) = 0.$$

Near a nondegenerate saddle,

$$V(t) = V(t_*) + \frac{1}{2}V''(t_*)(t - t_*)^2 + \dots$$

After the scaling $t - t_* = u/\sqrt{x}$, one obtains a sector

$$I_{t_*}(x) \sim e^{-xV(t_*)} x^{-1/2} \sum_{n \geq 0} c_n x^{-n}.$$

Each saddle supplies its own exponential monomial.

30.2 Several saddles produce a transseries

If saddles $t_1, \dots, t_r$ can contribute, the full asymptotic object has the schematic form

$$I(x) \sim \sum_{j=1}^r n_j e^{-xV(t_j)} x^{-1/2} \sum_{n \geq 0} c_{j,n} x^{-n}.$$

The coefficients n_j depend on how the original contour decomposes into steepest-descent cycles. As a parameter or direction changes, this decomposition can jump. That jump is the geometric origin of Stokes phenomena for integrals.

30.3 Relative actions

Choose one saddle as a reference. Then

$$e^{-xV(t_j)} = e^{-xV(t_0)} e^{-x(V(t_j) - V(t_0))}.$$

The differences

$$A_j = V(t_j) - V(t_0)$$

are the actions that appear as Borel singularity locations for the perturbative expansion around t_0 .

30.4 Why the coefficients diverge

The local expansion around one saddle has a finite region of analytic influence. High derivatives eventually feel neighboring saddles or singularities of V and a . This produces factorial growth. Steepest-descent geometry and Borel-plane resurgence are therefore two descriptions of the same global analytic structure.

30.5 Endpoint contributions

If the dominant point is an endpoint rather than a stationary point, the scaling may be $t = u/x$ rather than $u/\sqrt{x}$. This gives powers $x^{-1}, x^{-2}, \dots$. Degenerate saddles produce fractional powers, and coalescing saddles require uniform special-function models such as Airy or Pearcey functions.

30.6 Infinite-dimensional analogies

In quantum mechanics and quantum field theory, formal path integrals are expanded around classical solutions or saddles. The resulting perturbative series are commonly divergent, while instantons and other nonperturbative saddles produce factors like

$$e^{-A/g},$$

where g is a small coupling. Resurgent transseries provide a disciplined language for combining these sectors. Infinite-dimensional applications require substantial analytic care, but the finite-dimensional saddle picture explains the basic grammar.

Part VI

The wider algebraic landscape

Germes at infinity and Hardy fields

31.1 What is a germ?

Two functions $f, g : (a, \infty) \rightarrow \mathbb{R}$ define the same **germ at infinity** if they agree for all sufficiently large x . A germ forgets behavior on bounded intervals and remembers only the eventual function. This is the natural analytic setting for asymptotics.

Addition, multiplication, and differentiation are defined using representatives. For example, changing a function on $[0, 100]$ does not affect its germ at infinity.

31.2 Hardy fields

A **Hardy field** is a field of germs at infinity closed under differentiation. Classical examples are generated by rational functions, exponentials, and logarithms. Because a field cannot contain a nonzero germ with infinitely many zeros, every nonzero element has eventually constant sign. Hence Hardy fields are built for nonoscillatory asymptotics.

Example 31.1. The germs of

$$x, \quad \log x, \quad e^x, \quad x^{\sqrt{2}}(\log x)^{-3}, \quad \exp\left(\frac{x}{\log x}\right)$$

can belong to suitable Hardy fields. The germ of $\sin x$ cannot belong to any Hardy field: it has infinitely many zeros, so it would be a nonzero field element without a multiplicative inverse germ.

31.3 Orders of infinity become field comparisons

For positive germs f, g , write

$$f \prec g \iff f/g \rightarrow 0.$$

In a Hardy field, l'Hôpital-type principles often turn comparisons of functions into comparisons of logarithmic derivatives. This is the analytic ancestor of the valuation and asymptotic-couple structure in transseries.

31.4 Hardy's logarithmic-exponential functions

Starting from real constants and x , repeatedly apply field operations, exponentiation, and logarithm wherever defined. The resulting LE-functions form a Hardy field of germs. Examples include

$$\frac{\exp(\sqrt{x} + \log \log x)}{x^3 + \exp(x^{1/4})}.$$

Hardy studied their orders of infinity long before the modern formal transseries fields were constructed.

31.5 From a germ to a formal expansion

A germ may admit a transseries expansion obtained recursively by dominant comparison. For example,

$$\frac{1}{x + \log x} = \frac{1}{x} \left(1 + \frac{\log x}{x} \right)^{-1}$$

has expansion

$$x^{-1} - x^{-2} \log x + x^{-3} (\log x)^2 - \dots$$

The formal field provides a canonical arena in which such expansions can be added, multiplied, differentiated, and composed.

31.6 Embedding results

The field of Hardy's LE-germs embeds naturally into the field of LE-transseries, sending the germ of x to the transseries x and preserving order, field operations, exponentiation, and differentiation. This makes the formal objects faithful representatives for that tame analytic class [11, 12, 3].

Much broader Hardy fields can also be embedded into suitable extensions of transseries. Determining how far this relationship extends is an active structural theme: germs emphasize analytic existence, while transseries emphasize canonical formal normal forms.

31.7 Why germs and transseries are complementary

A germ has actual values but may lack a transparent normal form. A transseries has a transparent normal form but may need summation before it represents a function. The theory of H-fields abstracts the common differential-order behavior of both.

H-fields and asymptotic differential algebra

32.1 The basic axioms

An *H*-field is an ordered differential field K with constant field

$$C = \{c \in K : c' = 0\}$$

that satisfies two asymptotic principles. In a standard formulation:

1. if $f > C$, meaning f exceeds every constant, then $f' > 0$;
2. every bounded element has the form $c + \varepsilon$ with $c \in C$ and ε infinitesimal.

The second condition says that the valuation ring is $C + \mathfrak{o}$.

Both Hardy fields and the standard field of transseries are *H*-fields. The axioms capture eventual monotonicity and the existence of a constant “standard part” without mentioning functions or formal sums.

32.2 Asymptotic couples

Let $v(f)$ denote the valuation class of a nonzero element. The logarithmic derivative

$$f^\dagger = \frac{f'}{f}$$

depends asymptotically mainly on $v(f)$. This induces a map on the value group, often denoted

$$\psi(v(f)) = v(f^\dagger).$$

The pair consisting of the ordered value group and this derivative map is an **asymptotic couple**. It encodes the scale-changing behavior of differentiation.

For example,

$$(x^a)^\dagger = a/x, \quad (e^{x^2})^\dagger = 2x.$$

Differentiating a power lowers its magnitude, while differentiating a rapid exponential preserves its monomial and multiplies it by a large factor. The asymptotic couple organizes both regimes.

32.3 Liouville closure

A differential field is **Liouville closed** in the relevant ordered setting when it is real closed and has solutions to the basic first-order equations

$$y' = f, \quad \frac{z'}{z} = f \quad (z \neq 0).$$

The second equation amounts to exponentiating an antiderivative. The standard transseries field is Liouville closed. This explains, in structural language, why antiderivatives and exponential integrals remain inside the field.

32.4 Newtonianity

For an algebraic polynomial, Newton’s method cancels successive leading errors. Differential polynomials admit an analogous asymptotic Newton analysis. An H -field satisfying an appropriate differential Newton property is called **newtonian**. Informally, differential equations of Newton degree one have roots in the field.

The term should not be confused with numerical convergence of Newton iteration. It is a structural closure property expressed using valuations and differential polynomials.

32.5 The model-theoretic picture

Aschenbrenner, van den Dries, and van der Hoeven developed an axiomatic theory of transseries as H -fields. Conditions including Liouville closure, newtonianity, and an “omega-free” property isolate the first-order behavior of the standard transseries field. Their work yields strong completeness and model-theoretic transfer results [2, 3].

For a beginning reader, the philosophical consequence is more important than the formal logic:

Many theorems about transseries depend only on a small set of order, valuation, and differential axioms, not on the particular syntax used to construct the series.

32.6 Differential intermediate values

Ordinary real closed fields satisfy the intermediate value property for polynomials. Suitable transseries fields satisfy analogues for differential polynomials under sign-change hypotheses. Such results help explain why formal differential equations often have transseries solutions, while also clarifying the hypotheses needed for existence.

32.7 Small derivation

A derivation is called **small** when it sends infinitesimals or bounded elements into suitably controlled bounded elements. The exact valuation formulation depends on conventions. The familiar derivative on LE-transseries is small after a logarithmic change of independent variable in the settings where the model theory requires it. This technical adjustment aligns differentiation with the valuation topology.

Optional advanced viewpoint

The model theory does not merely classify one field. It provides a common language for Hardy fields, transseries, and surreal differential fields, and it suggests universal embedding and extension questions. This is why topics that look like formal asymptotics can lead naturally to valuation theory and logic.

Surreal numbers and transseries

33.1 A very brief introduction

Conway's surreal numbers form a proper class **No** containing the real numbers and all ordinal numbers, together with infinitely large and infinitesimal elements. Every surreal has a canonical normal form

$$a = \sum_{i < \alpha} r_i \omega^{a_i},$$

where α is an ordinal, $r_i \in \mathbb{R} \setminus \{0\}$, and the exponents a_i strictly decrease. This looks strikingly like a Hahn series with surreal exponents.

The simplest infinite ordinal ω plays a role analogous to the transseries variable x evaluated at an infinite value.

33.2 Exponential and logarithm

Gonshor constructed an exponential map

$$\exp : \mathbf{No} \rightarrow \mathbf{No}_{>0}$$

that extends the real exponential and has an inverse logarithm. Consequently, expressions such as

$$\exp(\omega), \quad \log \omega, \quad \exp(\sqrt{\omega} + \log \log \omega)$$

are genuine surreal numbers, not merely formal symbols.

33.3 Embedding transseries by sending x to ω

Standard fields of transseries can be embedded into **No** by sending

$$x \longmapsto \omega$$

and interpreting sums, exponentials, and logarithms in the surreal sense. A formal transseries then becomes a number of enormous or infinitesimal size. This makes precise the old intuition that a transseries is what an asymptotic expression would evaluate to at an “infinitely large x .”

33.4 A Hardy-type derivation on the surreals

Berarducci and Mantova constructed a derivation

$$\partial : \mathbf{No} \rightarrow \mathbf{No}$$

compatible with the field structure and exponential, with constant field $\mathbb{R}$, and satisfying Hardy-type order properties. With this derivation, the surreal numbers are Liouville closed; in particular, the derivation is surjective [5].

The basic normalization is consistent with

$$\partial(\omega) = 1,$$

so the embedded transseries variable behaves like an independent variable.

33.5 Omega-series

Inside $\mathbf{No}$, the **omega-series** form the smallest subfield containing $\mathbb{R}$ and ω and closed under exponential, logarithm, and all allowed summable families. They contain the familiar LE- and EL-series but are much larger. Berarducci and Mantova showed that omega-series can be composed and differentiated and can be interpreted as functions on positive infinite surreals [6].

33.6 Why not compose every surreal?

It is tempting to ask for a composition operation

$$a \circ b$$

for every surreal a and every positive infinite surreal b , compatible with the derivation and extending transseries composition. For the simplest surreal derivation, such a composition cannot exist on all of $\mathbf{No}$ with the desired compatibility. Omega-series form a large domain where composition works; the full surreal universe is too broad [6].

33.7 Numbers, functions, and formal expressions meet

A transseries can be viewed in three ways:

1. as a formal Hahn-like sum;
2. as an asymptotic germ after analytic realization;
3. as a surreal number after evaluation at an infinite argument.

These viewpoints are not identical, but the modern theory builds rigorous bridges among them.

Common pitfall

A surreal number is not automatically a function. Composition on omega-series supplies a functional interpretation for a large subclass, but it is incorrect to assume that every element of

No can be substituted into every other one.

LE-series, EL-series, omega-series, and hyper-series

34.1 Why several constructions exist

The phrase “close under exponentials, logarithms, and infinite sums” leaves choices about the order and extent of closure. Different choices produce different fields. They agree on a large common core but need not be isomorphic.

34.2 Logarithmic-exponential series

LE-series are built by a controlled hierarchy of finite exponential height and finite logarithmic depth, together with well-based or grid-based summation. This is the classical field used throughout most of this tutorial.

Informally, each individual LE-transseries can be produced from powers x^r by applying field operations, exponentiation, logarithm, and allowed infinite sums through finitely many nested construction stages.

34.3 Exponential-logarithmic series

EL-series arise from a different closure strategy based on prelogarithmic Hahn fields and repeated exponential extensions. Kuhlmann and collaborators developed these constructions in ordered exponential algebra.

Kuhlmann and Tressl proved that the LE-series field embeds as an exponential field into any of the relevant EL-series fields, but no non-Archimedean EL-series field embeds as an exponential field into the LE-series field. Thus EL-series are not merely a different notation for the same object [26].

34.4 Omega-series

Omega-series use the enormous summation resources of the surreal numbers. They contain LE- and EL-series and remain closed under composition and differentiation. Their supports and nesting complexity can extend far beyond any fixed finite-height construction.

34.5 Why hyperseries are considered

Even omega-series and familiar LE-series do not exhaust every desirable transfinite logarithmic scale. One may wish to represent iterates such as

$$\log_{\alpha} x$$

for transfinite indices α , or solve functional and differential equations whose natural supports pass through infinitely many logarithmic levels. **Hyperseries** and **logarithmic hyperseries** are frameworks designed for such transfinite iteration and summation [24].

34.6 A containment picture

The following diagram is schematic rather than a literal equality of every author's conventions:

Laurent/Puiseux series $\subset$ generalized power series $\subset$ LE-transseries $\subset$ omega-series $\subset$ **No**.

EL-series overlap the middle of this picture but are constructed differently and can contain elements not belonging to the LE field.

34.7 Grid-based and well-based variants revisited

At each level, one may impose a finitely generated grid or allow arbitrary reverse well-ordered support. Grid-based fields are more algorithmic; well-based fields have stronger closure and conceptual generality. Statements about composition, convergence, and fixed points can depend on which variant is chosen.

34.8 A comparison table

Framework	Main strength	Main caution
Poincaré series	Simple coefficient recurrences	Cannot see flat exponentials
Hahn series	Very general ordered supports	No exponential or derivation until added
LE-transseries	Rich differential, exponential, and compositional field	Finite construction height/depth per element
Resurgent transseries	Analytic summation and Stokes data	Special subclass; complex analytic hypotheses required
EL-series	Broad exponential closure of Hahn fields	Intrinsically different from LE-series
Omega-series	Huge surreal field with composition and derivation	Proper-class scale and deeper set-theoretic machinery
Hyperseries	Transfinite logarithmic/exponential iteration	Technically advanced and still developing

What transseries do not automatically solve

35.1 Oscillation

A totally ordered real transseries field cannot contain a nonzero element that changes sign forever. Functions such as

$$\sin x, \quad e^{-x} \sin(e^x)$$

require oscillatory extensions, complex phases, Fourier-type monomials, or other structures. One can develop such theories, but the simple dominance order must be modified.

35.2 Global information

An expansion at $x = +\infty$ does not determine values near $x = 0$, zeros on a bounded interval, global topology, or a definite integral involving the whole domain. Transseries are local to an asymptotic regime unless supplemented by analytic continuation or global equations.

35.3 Uniqueness of analytic realization

A formal transseries need not have a unique sum. Directional Borel sums can differ by Stokes jumps, and some formal series are not Borel summable. Stronger methods such as multisummation or accelero-summation may be required, and some series fall outside known summability classes.

35.4 Arbitrary rearrangement

Infinite transseries sums cannot be rearranged freely. Summability is a structural condition. A family that is summable in one grouping can become meaningless if a proposed rearrangement destroys well-based support or introduces infinitely many contributions to one coefficient.

35.5 Every differential equation

The transseries field solves many algebraic and differential equations, but not every imaginable equation has a solution in the smallest LE field. New exponential heights, iterated logarithms, hyperseries, oscillatory extensions, or analytic objects may be needed.

35.6 Convergence of formal composition

Formal composition is exact algebraically on its domain, but analytic substitution also requires that the represented functions and sectors overlap. A formal identity can outlive the numerical region in which a particular truncation is useful.

35.7 Automatic numerical accuracy

A transseries with ten displayed terms is not necessarily more accurate than one with five. Divergent sectors should be optimally truncated. Exponentially small sectors can be important near Stokes lines or boundary layers even when they are negligible elsewhere.

35.8 A diagnostic checklist

Before using a transseries answer, ask:

1. What is the limiting variable and direction?
2. Which monomial ordering is intended?
3. Is the support legal in the chosen framework?
4. Is the statement formal, asymptotic, or summability-based?
5. What is the first omitted scale?
6. Are there Stokes directions or boundary layers?
7. Which constants or transseries parameters are fixed by the problem?

A guided reading map

36.1 For a first conceptual pass

Begin with Edgar’s *Transseries for Beginners*, which develops grid-based and well-based monomials, examples, differentiation, composition, and convergence with unusually concrete explanations [17]. The present tutorial was designed to make that entry point still more elementary.

36.2 For formal algebra and computation

Van der Hoeven’s monograph *Transseries and Real Differential Algebra* gives a systematic computational and differential-algebraic treatment [23]. The papers of van den Dries, Macintyre, and Marker construct the LE field and establish its model-theoretic and analytic foundations [11, 12].

36.3 For H -fields and model theory

The monograph by Aschenbrenner, van den Dries, and van der Hoeven is the central modern reference for asymptotic differential algebra and the model theory of transseries [2]. Their expository article on numbers, germs, and transseries is a shorter orientation [3].

36.4 For Borel summation and resurgence

Costin’s book develops Borel summability and nonlinear differential equations rigorously [10]. Dorigoni’s introduction and the Physics Reports primer by Aniceto, Bařar, and Schiappa provide broad, calculation-rich introductions to resurgence and physical applications [15, 1].

36.5 For surreal numbers

Conway and Gonshor are the classical references for surreal arithmetic and exponentiation [8, 20]. Berarducci and Mantova construct the Hardy-type surreal derivation and then develop omega-series, composition, and surreal analytic functions [5, 6].

36.6 For composition and iteration

Edgar’s papers on composition and fractional iteration are direct continuations of the beginner article and include careful support control in well-based and grid-based settings [18, 19].

36.7 For the historical roots

Hahn’s generalized series, Hardy’s orders of infinity, Borel’s summation theory, and Ècalle’s analyzable functions are four major roots of the subject [21, 22, 7, 16]. Modern transseries bring these roots into a common language.

Checkpoint

A sensible learning order is:

asymptotic scales $\longrightarrow$ Hahn series $\longrightarrow$ LE operations
 $\longrightarrow$ ODE examples $\longrightarrow$ Borel–Stokes analysis
 $\longrightarrow$ H -fields and surreals.

Trying to begin with alien calculus or model theory before mastering dominant monomials usually makes the subject look harder than it is.

Part VII

Exercises, solutions, and reference material

Exercises

The exercises are grouped by topic. A star indicates a problem that is longer or more conceptual, not one that requires advanced prerequisites. Complete solutions begin in Chapter 38.

37.1 Asymptotic scales and support

Exercise 37.1. Arrange the following monomials from largest to smallest as $x \rightarrow +\infty$:

$$x^{-10}, \quad e^{-\sqrt{x}}, \quad (\log x)^{100}x^{-3}, \quad e^{-x}, \quad x^2e^{-\sqrt{x}}, \quad e^{-x/\log x}.$$

Justify every neighboring comparison.

Exercise 37.2. Show directly from the definition that

$$\log x = o(x^a)$$

for every $a > 0$, and that

$$x^b = o(e^{x^a})$$

for every $a, b > 0$.

Exercise 37.3. For

$$F(x) = \int_0^\infty \frac{e^{-xp}}{1+p} dp,$$

perform N integrations by parts and prove that

$$F(x) \sim_{\text{as}} \sum_{n \geq 0} (-1)^n n! x^{-n-1}.$$

Give an explicit bound for the remainder when $x > 0$.

Exercise 37.4. Decide which of the following supports are legal at $+\infty$ under the reverse-well-ordered convention:

$$S_1 = \{x^{-n} : n \in \mathbb{N}\},$$

$$S_2 = \{x^n : n \in \mathbb{N}\},$$

$$S_3 = \{e^{-nx} : n \in \mathbb{N}\},$$

$$S_4 = \{x^{-1/n} : n \geq 1\},$$

$$S_5 = \{x^{-n}e^{-mx} : n, m \in \mathbb{N}\}.$$

For each illegal support, exhibit a nonempty subset with no largest element.

Exercise 37.5. Let

$$A = 3x^2 - 5x + e^{-x}, \quad B = 3x^2 - 5x + x^{-7}.$$

Determine $\text{mag}(A)$, $\text{dom}(A)$, $\text{mag}(A - B)$, and whether $A \sim B$.

Exercise 37.6 (Hahn-product finiteness). Let S, T be reverse well-ordered subsets of an ordered abelian group. Explain why a fixed group element g can have only finitely many factorizations $g = st$ with $s \in S$ and $t \in T$. You may argue by contradiction using monotone subsequences.

37.2 Algebra, exponentials, and logarithms

Exercise 37.7. Expand

$$\frac{1}{x + 1 + e^{-x}}$$

through algebraic order x^{-4} and through the first exponential sector down to $x^{-3}e^{-x}$.

Exercise 37.8. Compute the displayed terms of

$$\log \left(2x^3 e^{\sqrt{x}} \left(1 - \frac{1}{x} + 2e^{-x} \right) \right)$$

through algebraic order x^{-3} and through terms of order $x^{-2}e^{-x}$.

Exercise 37.9. Expand

$$(x^2 + x + e^{-x})^{3/2}$$

through algebraic order x^{-1} and include the leading exponential correction.

Exercise 37.10. Let

$$T = x + 2 \log x + x^{-1}.$$

Expand e^{-T} through order $x^{-4}e^{-x}$.

Exercise 37.11. Show that if $A \succ B > 0$, then

$$A + B \sim A, \quad \log(A + B) = \log A + \frac{B}{A} + O((B/A)^2).$$

State the formal support condition under which the error expansion is meaningful.

37.3 Differentiation, integration, and composition

Exercise 37.12. Compute the logarithmic derivative of

$$\mathfrak{m} = x^a (\log x)^b \exp(-x + \sqrt{x}).$$

Use it to find the dominant term of $\mathfrak{m}'$.

Exercise 37.13. Find coefficients b_0, b_1, b_2, b_3 such that

$$\frac{d}{dx} \left[e^{x^3} (b_0 x^{-2} + b_1 x^{-5} + b_2 x^{-8} + b_3 x^{-11}) \right] = e^{x^3} + O(e^{x^3} x^{-12}).$$

Exercise 37.14. Expand

$$\log(x^2 + x + e^{-x})$$

through algebraic order x^{-4} and through the leading exponential term.

Exercise 37.15. Let $F(x) = x + \log x$. Derive the first three correction terms in the compositional inverse

$$F^{[-1]}(x) = x - \log x + \cdots.$$

Verify by substitution through order x^{-2} .

Exercise 37.16. Prove formally that, when $U \prec G$ and the Taylor family is summable,

$$e^{G+U} = e^G \sum_{n \geq 0} \frac{U^n}{n!}.$$

Explain why the same formula is not used with $U = G = x$ at infinity.

37.4 Special functions and equations

Exercise 37.17. Starting from $W + \log W = \log x$, derive

$$W = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} + O\left(\frac{L_2^3}{L_1^3}\right).$$

Exercise 37.18. For $y' + xy = 1$, seek

$$y \sim \sum_{n \geq 0} a_n x^{-2n-1}.$$

Find the recurrence, solve it, and identify the exponentially small homogeneous solution omitted by the power sector.

Exercise 37.19. Use Section 20.2 to compute $a_1, \dots, a_5$ for Section 20.1. Then derive the leading factor $e^{-x}x^2$ of the first nonperturbative sector from the linearized equation.

Exercise 37.20. For the Airy Riccati equation $u' + u^2 = x$, take

$$u = -x^{1/2} + c_1 x^{-1} + c_2 x^{-5/2} + c_3 x^{-4} + \cdots.$$

Find c_1, c_2, c_3 . Integrate to recover the leading exponential and algebraic prefactor of the decaying Airy solution.

Exercise 37.21. Solve Section 22.3 directly and identify which terms are visible in the outer expansion for fixed $0 < x < 1$. What changes when $x = \varepsilon X$?

Exercise 37.22. Apply the operator expansion

$$\frac{D}{e^D - 1} = 1 - \frac{D}{2} + \frac{D^2}{12} - \frac{D^4}{720} + \cdots$$

to $\log x$ and integrate to recover Stirling's series through x^{-3} .

Exercise 37.23 (Abel coordinate). For $T(x) = x + x^{-1}$, substitute

$$A(x) = \frac{x^2}{2} + c \log x + ax^{-2} + O(x^{-4})$$

into $A(T) - A = 1$. Determine c and a .

Exercise 37.24. Use Equation (24.2) to write $T^{[1/2]}(x)$ through x^{-7} . Check formally that composing this half-iterate with itself gives $x + x^{-1} + O(x^{-9})$.

37.5 Borel summation and resurgence

Exercise 37.25. Compute the Borel transform and Borel sum of

$$\sum_{n \geq 0} (-1)^n n! x^{-n-1}.$$

Verify the first three asymptotic coefficients by integration by parts.

Exercise 37.26. For

$$\hat{\phi}(p) = \frac{1}{(1 - p/A)^2},$$

find the coefficients a_n of the original formal series

$$\tilde{\phi}(x) = \sum_{n \geq 0} a_n x^{-n-1}.$$

What exponential scale is predicted by the Borel singularity?

Exercise 37.27. Compute the difference between upper and lower lateral Laplace integrals of

$$\frac{1}{1 - p/A}, \quad A > 0,$$

along the positive ray. State clearly your orientation convention.

Exercise 37.28. Show that the Borel transform turns the product

$$x^{-1} \cdot x^{-1} = x^{-2}$$

into a convolution, using the convention $\mathcal{B}(x^{-n-1}) = p^n/n!$.

Exercise 37.29. Suppose

$$a_n \sim CA^{-n}\Gamma(n + \beta), \quad A > 0.$$

Use the ratio of successive terms to estimate the optimal truncation index for $\sum a_n x^{-n}$. Use Stirling's formula to identify the exponential scale of the least term.

Exercise 37.30 (Large-order extraction). Assume exact data

$$a_n = S \frac{\Gamma(n + \beta)}{2\pi i A^{n+\beta}} \left(b_0 + \frac{Ab_1}{n + \beta - 1} \right).$$

Construct a sequence that tends to $Sb_0/(2\pi i)$ as $n \rightarrow \infty$, and a second rescaled difference that tends to $Sb_1/(2\pi i)$.

37.6 Structural questions

Exercise 37.31. Explain why a Hardy field cannot contain the germ of $\sin x$. Which field axiom fails if one tries to adjoin it?

Exercise 37.32. Give an example of two distinct functions with the same complete inverse-power asymptotic expansion at $+\infty$. Then give a transseries scale that distinguishes them.

Exercise 37.33. Explain the difference among the following three claims about a formal expression: it solves an equation formally; it is asymptotic to a function; it is Borel summable to a function.

Exercise 37.34. Why does sending x to the surreal number ω make x^{-1} infinitesimal and e^x larger than every power x^n ? Express the answer using order-preserving field and exponential operations.

Exercise 37.35. Make a two-column comparison of grid-based and well-based transseries. Include support, computational convenience, and closure/generalization.

Exercise 37.36 (Designing an ansatz). A nonlinear equation has a perturbative power series and its linearization has two independent small solutions

$$e^{-Ax}x^\alpha, \quad e^{-Bx}x^\beta,$$

where $A, B > 0$ are rationally independent. Write a natural two-parameter transseries ansatz and describe the action lattice generated by nonlinear products.

Exercise 37.37 (Residual certification). Suppose a first-order equation $E[y] = 0$ has linearization L at a formal solution and an approximation y_N satisfies

$$E[y_N] = r\mathfrak{m} + \text{smaller terms.}$$

Explain how to choose a correction cn from the dominant relation $L[cn] \sim -r\mathfrak{m}$. What can go wrong if $L[\mathfrak{n}]$ is anomalously small?

Exercise 37.38 (Saddles). For the Gaussian integral

$$I(x) = \int_{-\infty}^{\infty} e^{-xt^2/2} dt,$$

derive its saddle scaling and evaluate it exactly. Then explain why a nonquadratic analytic potential generally produces an infinite inverse-power prefactor series.

Solutions to the exercises

38.1 Asymptotic scales and support

Solution 1

The order is

$$(\log x)^{100} x^{-3} \succ x^{-10} \succ e^{-\sqrt{x}} x^2 \succ e^{-\sqrt{x}} \succ e^{-x/\log x} \succ e^{-x}.$$

The first comparison follows because $(\log x)^{100} x^7 \rightarrow \infty$. Any inverse power dominates $x^2 e^{-\sqrt{x}}$ because exponentials beat powers. Multiplying by x^2 makes $x^2 e^{-\sqrt{x}}$ larger than $e^{-\sqrt{x}}$. Since

$$\frac{x}{\log x} \gg \sqrt{x},$$

we have $e^{-\sqrt{x}} \gg e^{-x/\log x}$. Finally $x/\log x < x$ for large x , so $e^{-x/\log x} \gg e^{-x}$.

Solution 2

For the first limit, repeated l'Hôpital gives

$$\lim_{x \rightarrow \infty} \frac{\log x}{x^a} = \lim_{x \rightarrow \infty} \frac{x^{-1}}{ax^{a-1}} = \lim_{x \rightarrow \infty} \frac{1}{ax^a} = 0.$$

For the second, take logarithms of the positive ratio:

$$\log \frac{x^b}{e^{x^a}} = b \log x - x^a \rightarrow -\infty$$

by the first result. Hence the ratio tends to zero.

Solution 3

Let

$$F_k(x) = \int_0^\infty \frac{e^{-xp}}{(1+p)^{k+1}} dp.$$

Integrating by parts with $u = (1+p)^{-k-1}$ and $dv = e^{-xp} dp$ gives

$$F_k = \frac{1}{x} - \frac{k+1}{x} F_{k+1}.$$

Iteration yields

$$F(x) = \sum_{n=0}^{N-1} (-1)^n n! x^{-n-1} + (-1)^N N! x^{-N} F_N(x).$$

Since $0 < F_N(x) \leq \int_0^\infty e^{-xp} dp = x^{-1}$,

$$|R_N(x)| \leq N! x^{-N-1}.$$

This proves the asymptotic expansion.

Solution 4

S_1, S_3 , and S_5 are legal. Every nonempty subset has a largest monomial because decreasing an exponent n or m increases the monomial, and the relevant set of integer pairs has a lexicographically minimal element. The support S_2 is illegal: the whole set has no largest element. The support S_4 is illegal because

$$x^{-1} \prec x^{-1/2} \prec x^{-1/3} \prec \dots$$

has no largest member.

Solution 5

Both A and B have magnitude x^2 and dominant term $3x^2$. Their difference is

$$A - B = e^{-x} - x^{-7} \sim -x^{-7},$$

so $\text{mag}(A - B) = x^{-7}$. Since $A - B \prec A$, we have $A \sim B$.

Solution 6

Assume infinitely many factorizations $g = s_n t_n$. After passing to a subsequence, reverse well-ordering lets us take $s_1 \prec s_2 \prec \dots$ or an analogous strict monotone pattern among distinct s_n . Then

$$t_n = g/s_n$$

moves strictly in the opposite direction, producing an infinite increasing sequence in T with no largest element. This contradicts reverse well-ordering. A precise proof uses the fact that every infinite subset of a reverse well-ordered set has a monotone subsequence whose forbidden direction cannot continue indefinitely.

38.2 Algebra, exponentials, and logarithms

Solution 7

Factor x and set $u = x^{-1} + x^{-1}e^{-x}$:

$$\frac{1}{x + 1 + e^{-x}} = x^{-1}(1 + u)^{-1}.$$

Expanding and retaining the requested scales gives

$$\begin{aligned} \frac{1}{x + 1 + e^{-x}} &= x^{-1} - x^{-2} + x^{-3} - x^{-4} \\ &\quad - x^{-2}e^{-x} + 2x^{-3}e^{-x} + \text{smaller terms.} \end{aligned}$$

Solution 8

Let $u = -x^{-1} + 2e^{-x}$. Then

$$\log T = \log 2 + 3 \log x + \sqrt{x} + \log(1 + u).$$

Using $u - u^2/2 + u^3/3$ gives

$$\begin{aligned} \log T &= \sqrt{x} + 3 \log x + \log 2 - x^{-1} - \frac{1}{2}x^{-2} - \frac{1}{3}x^{-3} \\ &\quad + 2e^{-x} + 2x^{-1}e^{-x} + 2x^{-2}e^{-x} + \text{smaller terms.} \end{aligned}$$

Solution 9

Factor x^2 :

$$(x^2 + x + e^{-x})^{3/2} = x^3(1 + x^{-1} + x^{-2}e^{-x})^{3/2}.$$

The binomial coefficients are $1, 3/2, 3/8, -1/16, 3/128, \dots$. Hence

$$x^3 + \frac{3}{2}x^2 + \frac{3}{8}x - \frac{1}{16} + \frac{3}{128x} + \frac{3}{2}xe^{-x} + \text{smaller terms.}$$

Cross terms with the exponential are smaller than the leading xe^{-x} term.

Solution 10

We have

$$e^{-T} = e^{-x}x^{-2}e^{-x^{-1}}.$$

Therefore

$$e^{-T} = e^{-x} \left(x^{-2} - x^{-3} + \frac{1}{2}x^{-4} + O(x^{-5}) \right).$$

Solution 11

Write $A + B = A(1 + B/A)$. Since $B/A \prec 1$,

$$A + B \sim A$$

and

$$\log(A + B) = \log A + \log(1 + B/A) = \log A + B/A + O((B/A)^2).$$

The family of powers $(B/A)^n$ must be summable; this holds in the standard Hahn setting when B/A is infinitesimal with well-based support.

38.3 Differentiation, integration, and composition**Solution 12**

Taking a logarithm,

$$\log m = a \log x + b \log \log x - x + \sqrt{x}.$$

Thus

$$\mathfrak{m}^\dagger = -1 + \frac{1}{2\sqrt{x}} + \frac{a}{x} + \frac{b}{x \log x}.$$

The parenthesis is asymptotic to -1 , so

$$\mathfrak{m}' \sim -\mathfrak{m}.$$

Solution 13

Let $B = b_0x^{-2} + b_1x^{-5} + b_2x^{-8} + b_3x^{-11}$. We need

$$B' + 3x^2B = 1 + O(x^{-12}).$$

Matching powers gives

$$3b_0 = 1, \quad 3b_1 - 2b_0 = 0, \quad 3b_2 - 5b_1 = 0, \quad 3b_3 - 8b_2 = 0.$$

Hence

$$b_0 = \frac{1}{3}, \quad b_1 = \frac{2}{9}, \quad b_2 = \frac{10}{27}, \quad b_3 = \frac{80}{81}.$$

Solution 14

Factor x^2 and set $u = x^{-1} + x^{-2}e^{-x}$. Then

$$\begin{aligned} \log(x^2 + x + e^{-x}) &= 2\log x + x^{-1} - \frac{1}{2}x^{-2} + \frac{1}{3}x^{-3} - \frac{1}{4}x^{-4} \\ &\quad + x^{-2}e^{-x} + \text{smaller terms.} \end{aligned}$$

Solution 15

Write $G = x - L + a/x + b/x^2$ with $L = \log x$ and allow a, b to depend polynomially on L . Expanding

$$\log G = L + \log \left(1 - \frac{L}{x} + \frac{a}{x^2} + \frac{b}{x^3} \right)$$

gives, through x^{-2} ,

$$F(G) - x = \frac{a - L}{x} + \frac{b + a - L^2/2}{x^2} + \dots.$$

Thus $a = L$ and $b = L^2/2 - L$. Therefore

$$F^{[-1]}(x) = x - L + \frac{L}{x} + \frac{L^2/2 - L}{x^2} + \dots.$$

Substitution makes the residual $O(L^3/x^3)$.

Solution 16

The additive decomposition regards G as the large part and U as infinitesimal relative to the chosen Taylor scale. By the definition of transseries exponentiation,

$$e^{G+U} = e^G e^U = e^G \sum_{n \geq 0} U^n / n!.$$

For $U = x$, the powers $1, x, x^2, \dots$ have no largest term and are not a legal decreasing support at infinity; e^x must be treated as one monomial.

38.4 Special functions and equations

Solution 17

Set $W = L_1 - L_2 + u$. The reduced equation is

$$u + \log \left(1 - \frac{L_2 - u}{L_1} \right) = 0.$$

Assume $u = P_1/L_1 + P_2/L_1^2 + \dots$. At order L_1^{-1} , $P_1 = L_2$. At order L_1^{-2} ,

$$P_2 = L_2^2/2 - L_2.$$

This gives the stated expansion.

Solution 18

Substitution into $y' + xy = 1$ gives

$$a_0 = 1, \quad a_{n+1} = (2n+1)a_n.$$

Hence $a_n = (2n-1)!!$. The homogeneous equation $y' + xy = 0$ has solution

$$Ce^{-x^2/2},$$

which is smaller than every inverse power and therefore absent from the algebraic sector.

Solution 19

The recurrence gives

$$a_1 = 1, \quad a_2 = 2, \quad a_3 = 8, \quad a_4 = 44, \quad a_5 = 296.$$

The linearized perturbation obeys

$$\delta' + (1 - 2y_0)\delta = 0.$$

Thus

$$\delta = C \exp \left(-x + 2 \int y_0 dx \right) = Ce^{-x} x^2 (1 + o(1)).$$

Solution 20

Insert the ansatz into $u' + u^2 = x$. Matching $x^{-1/2}$ gives

$$-\frac{1}{2} - 2c_1 = 0, \quad c_1 = -\frac{1}{4}.$$

Matching x^{-2} gives

$$-c_1 + c_1^2 - 2c_2 = 0, \quad c_2 = \frac{5}{32}.$$

Matching $x^{-7/2}$ gives

$$-\frac{5}{2}c_2 + 2c_1c_2 - 2c_3 = 0, \quad c_3 = -\frac{15}{64}.$$

Integrating begins with

$$\log y = -\frac{2}{3}x^{3/2} - \frac{1}{4}\log x + \dots,$$

so

$$y \asymp x^{-1/4} e^{-2x^{3/2}/3}.$$

Solution 21

Solving once gives $y' = Ce^{-x/\varepsilon}$ and then

$$y = A + Be^{-x/\varepsilon}.$$

The boundary conditions yield

$$y = \frac{1 - e^{-x/\varepsilon}}{1 - e^{-1/\varepsilon}}.$$

For fixed $0 < x < 1$, both exponentials are beyond all powers and the outer expansion is $y \sim 1$. If $x = \varepsilon X$, then $e^{-x/\varepsilon} = e^{-X}$ is order one and resolves the layer at $x = 0$.

Solution 22

Applying the operator to $\log x$ gives

$$F' = \log x - \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} + \cdots.$$

Integrating,

$$F = \left(x - \frac{1}{2}\right) \log x - x + C + \frac{1}{12x} - \frac{1}{360x^3} + \cdots.$$

For $F = \log \Gamma$, $C = \frac{1}{2} \log(2\pi)$.

Solution 23

Use

$$T^2 = x^2 + 2 + x^{-2}, \quad \log T = \log x + \log(1 + x^{-2}).$$

The constant term in $A(T) - A$ is already 1. Since

$$\log(T/x) = \log(1 + x^{-2}) = x^{-2} - \frac{1}{2}x^{-4} + \cdots$$

and

$$T^{-2} - x^{-2} = -2x^{-4} + \cdots,$$

the coefficient of x^{-2} is $1/2 + c$, so $c = -1/2$. The coefficient of x^{-4} is $-c/2 - 2a$, so $a = 1/8$. Thus

$$A = \frac{x^2}{2} - \frac{1}{2} \log x + \frac{1}{8x^2} + \cdots.$$

Solution 24

Set $s = 1/2$ in Equation (24.2):

$$T^{[1/2]}(x) = x + \frac{1}{2x} + \frac{1}{8x^3} + \frac{1}{16x^5} + \frac{3}{128x^7} + O(x^{-9}).$$

Taylor-composing this expression with itself and collecting powers gives

$$x + x^{-1} + O(x^{-9}).$$

The group law of the Abel construction guarantees the cancellation; direct expansion verifies it coefficient by coefficient.

38.5 Borel summation and resurgence

Solution 25

The Borel transform is

$$\sum_{n \geq 0} (-p)^n = \frac{1}{1+p}.$$

Therefore

$$\mathcal{S}_0 \tilde{F} = \int_0^\infty \frac{e^{-xp}}{1+p} dp = -e^x \operatorname{Ei}(-x).$$

Three integrations by parts give

$$-e^x \operatorname{Ei}(-x) = x^{-1} - x^{-2} + 2x^{-3} + O(x^{-4}),$$

matching $0!, -1!, 2!$ with alternating signs.

Solution 26

Since

$$(1-z)^{-2} = \sum_{n \geq 0} (n+1)z^n,$$

we have

$$\hat{\phi}(p) = \sum_{n \geq 0} (n+1)A^{-n}p^n.$$

Thus

$$a_n = n!(n+1)A^{-n} = (n+1)!A^{-n}.$$

The singularity at $p = A$ predicts the exponential scale e^{-Ax} .

Solution 27

The integrand is

$$\frac{e^{-xp}}{1-p/A}.$$

Its residue at $p = A$ is $-Ae^{-Ax}$. With upper contour from 0 to $+\infty$ and lower contour subtracted in the same direction, the resulting closed contour is clockwise, so

$$\mathcal{S}_{0+} - \mathcal{S}_{0-} = -2\pi i(-Ae^{-Ax}) = 2\pi iAe^{-Ax}.$$

Changing the convention reverses the sign.

Solution 28

We have

$$\mathcal{B}(x^{-1}) = 1.$$

The convolution is

$$(1 * 1)(p) = \int_0^p 1 dq = p.$$

But

$$\mathcal{B}(x^{-2}) = p,$$

so the rule holds in this simplest case.

Solution 29

The successive-term ratio is approximately

$$\frac{n + \beta}{Ax}.$$

It crosses one near $n \approx Ax - \beta$, so the optimal index is $N \approx Ax$. Stirling's formula then gives a least-term scale

$$x^{\beta-1/2} e^{-Ax}$$

up to a constant and convention-dependent power of A .

Solution 30

Define

$$R_n := \frac{A^{n+\beta} a_n}{\Gamma(n + \beta)}.$$

Then

$$R_n = \frac{S}{2\pi i} \left(b_0 + \frac{Ab_1}{n + \beta - 1} \right) \longrightarrow \frac{Sb_0}{2\pi i}.$$

Let $L = \lim R_n$. Then

$$\frac{n + \beta - 1}{A} (R_n - L) \longrightarrow \frac{Sb_1}{2\pi i}.$$

With finite data, one first extrapolates R_n to estimate L and then applies the second rescaling.

38.6 Structural questions**Solution 31**

The germ of $\sin x$ has infinitely many zeros but is not eventually zero. In a field, every nonzero element must have a multiplicative inverse. The reciprocal $1/\sin x$ is not a germ of a function defined on any entire tail interval because poles occur arbitrarily far out. Hence field closure fails.

Solution 32

The functions 0 and e^{-x} have the same complete inverse-power expansion, namely zero. The enlarged scale

$$1, x^{-1}, x^{-2}, \dots, e^{-x}, x^{-1}e^{-x}, \dots$$

distinguishes them at the e^{-x} monomial.

Solution 33

A formal solution means substitution produces the zero formal transseries. An asymptotic representation means an actual function has remainders controlled by every finite truncation on a stated domain. Borel summability is stronger and more specific: after Borel continuation and a directional Laplace transform, a prescribed analytic function is reconstructed from the formal coefficients.

Solution 34

An order-preserving embedding sends $x > n$ for every integer n to $\omega > n$. Taking reciprocals reverses positive inequalities, so

$$0 < \omega^{-1} < 1/n$$

for every n . Because the exponential is order preserving and grows faster than powers in the embedded transseries field,

$$\exp(\omega) > \omega^n$$

for every fixed n .

Solution 35

Grid-based supports lie in a finitely generated multiplicative grid, which makes exponents reducible to integer multi-indices and supports effective computation. Well-based supports need only be reverse well ordered; they include more series and often have stronger conceptual closure, but can require subtler support arguments and may be less directly algorithmic.

Solution 36

Use two parameters C_1, C_2 :

$$y = \sum_{k, \ell \geq 0} C_1^k C_2^\ell e^{-(kA + \ell B)x} x^{k\alpha + \ell\beta} \Phi_{k, \ell}(x).$$

The action lattice is

$$\{kA + \ell B : k, \ell \in \mathbb{N}\}.$$

Rational independence prevents accidental equality between distinct pairs, while nonlinear products add the index pairs.

Solution 37

Compute the dominant effect of L on candidate monomials. If

$$L[\mathbf{n}] \sim \lambda \mathbf{m},$$

choose $c = -r/\lambda$. The new residual is smaller. If $L[\mathbf{n}]$ is anomalously small because of resonance or because $\mathbf{n}$ lies near the kernel, division by the dominant coefficient fails. One may need a logarithmic factor, a different monomial, or a new free homogeneous sector.

Solution 38

Set $u = \sqrt{x}t$. Then

$$I(x) = x^{-1/2} \int_{-\infty}^{\infty} e^{-u^2/2} du = \sqrt{\frac{2\pi}{x}}.$$

The potential is exactly quadratic, so the saddle expansion terminates. For a nonquadratic analytic V , Taylor terms $V^{(k)}(0)t^k$ become powers of $x^{-1/2}$ after rescaling; Gaussian moments generate an infinite asymptotic series, usually factorially divergent because neighboring saddles or singularities control late orders.

A proof kit for formal asymptotics

This appendix collects short arguments that recur throughout the book. They are not a substitute for the full support theorems in the specialist literature, but they explain why the elementary manipulations work and exactly where the nontrivial hypotheses enter.

A.1 Uniqueness of coefficients in an asymptotic scale

Let $\phi_0, \phi_1, \dots$ be nonzero functions on a tail interval, with

$$\phi_{n+1} = o(\phi_n) \quad (x \rightarrow +\infty).$$

Suppose a function f has two asymptotic expansions in this same scale,

$$f \sim \sum_{n \geq 0} a_n \phi_n, \quad f \sim \sum_{n \geq 0} b_n \phi_n.$$

Then $a_n = b_n$ for every n .

Proof. If the coefficients differ, let k be the first index with $a_k \neq b_k$. Subtract the two expansions through index k :

$$0 = (a_k - b_k)\phi_k + o(\phi_k).$$

Divide by ϕ_k . The left side remains zero, while the right side tends to $a_k - b_k \neq 0$, a contradiction. $\square$

Intuition

Asymptotic coefficients are read from left to right. Once all larger scales have been removed, the first surviving scale cannot be canceled by smaller ones.

A.2 Why exponentials beat powers

For every fixed $a > 0$ and every real N ,

$$e^{-ax} = o(x^{-N}) \quad (x \rightarrow +\infty).$$

Indeed,

$$\log(x^N e^{-ax}) = N \log x - ax \longrightarrow -\infty,$$

because $\log x/x \rightarrow 0$. Exponentiating gives $x^N e^{-ax} \rightarrow 0$. Replacing x by x^α with $\alpha > 0$ gives the more general comparison

$$e^{-ax^\alpha} = o(x^{-N}).$$

This elementary fact is the mathematical reason that an exponential correction is invisible to every finite inverse-power expansion.

A.3 Leading monomials and inverses

Let

$$T = a\mathbf{m}(1 + \varepsilon), \quad a \neq 0, \quad \varepsilon \prec 1,$$

where $\mathbf{m}$ is a monomial and every monomial in ε is smaller than 1. Formally,

$$T^{-1} = a^{-1}\mathbf{m}^{-1}(1 + \varepsilon)^{-1} = a^{-1}\mathbf{m}^{-1} \sum_{n \geq 0} (-\varepsilon)^n.$$

The algebraic identity is just the geometric series. The support issue is subtler: for any fixed output monomial, only finitely many products from the powers ε^n may contribute. A well-based or grid-based support condition guarantees this local finiteness.

Proposition A.1 (geometric inversion in a finitely generated grid). *Let $\boldsymbol{\mu} = (\mu_1, \dots, \mu_r)$ with each $\mu_i \prec 1$, and suppose every monomial of ε belongs to*

$$\boldsymbol{\mu}^{\mathbb{N}^r} = \left\{ \mu_1^{k_1} \cdots \mu_r^{k_r} : k_i \in \mathbb{N} \right\}$$

and has positive total degree. Then the coefficient of any given grid monomial in $\sum_{n \geq 0} (-\varepsilon)^n$ receives contributions from only finitely many n .

Proof. A monomial occurring in ε^n has total multi-index degree at least n . A fixed target multi-index has finite total degree, so it cannot occur once n exceeds that degree. $\square$

A.4 Formal exponentials and logarithms

If $S \prec 1$, define

$$\exp(S) = \sum_{n \geq 0} \frac{S^n}{n!}, \quad \log(1 + S) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} S^n.$$

In a formal transseries field, these formulas do not assert numerical convergence. They assert that the families of supports are summable: each monomial receives a well-defined coefficient, and the union of all resulting supports remains legal.

For a general positive transseries, first factor its dominant monomial:

$$T = a\mathbf{m}(1 + S), \quad a > 0, \quad S \prec 1.$$

Then

$$\log T = \log a + \log \mathbf{m} + \log(1 + S).$$

Likewise, split a general exponent into its purely large, constant, and small parts,

$$U = U_{\succ 1} + c + U_{\prec 1},$$

and write

$$e^U = e^c e^{U_{\succ 1}} e^{U_{\prec 1}}.$$

The first two factors determine a new monomial; only the last is expanded as a Taylor series.

A.5 A one-step Newton principle

Suppose we seek y with $F(y) = 0$, and an approximation y_0 leaves residual

$$r = F(y_0).$$

Let $L = F'[y_0]$ denote the linearized operator. Choose a correction δ satisfying, to dominant order,

$$L\delta \sim -r.$$

Taylor expansion gives

$$F(y_0 + \delta) = r + L\delta + Q(\delta),$$

where Q is at least quadratic. If the chosen δ cancels the dominant monomial of r and if $Q(\delta)$ is smaller, the new residual has strictly smaller magnitude. Repetition produces a transseries coefficient by coefficient.

Common pitfall

The step can fail at a resonance. If the dominant coefficient of L vanishes on the candidate monomial, the correction may require a logarithm, a changed power, or a new free homogeneous sector. Resonance is information, not merely an algebraic inconvenience.

A.6 Why formal differentiation is usually easy

For a monomial

$$\mathfrak{m} = x^a (\log x)^b \exp(L),$$

logarithmic differentiation gives

$$\frac{\mathfrak{m}'}{\mathfrak{m}} = \frac{a}{x} + \frac{b}{x \log x} + L'.$$

Thus

$$\mathfrak{m}' = \mathfrak{m} \left(\frac{a}{x} + \frac{b}{x \log x} + L' \right).$$

A transseries derivative is obtained termwise, provided the differentiated family remains summable. In the standard LE fields this closure is built into the construction. For arbitrary generalized series, however, termwise differentiation is not automatic: a derivative could move infinitely many terms onto the same monomial or destroy well-basedness.

A.7 Asymptotic validity of algebraic operations

Suppose actual functions f and g have asymptotic expansions in compatible scales and nonzero leading terms. Then finite truncation estimates immediately justify addition and multiplication. For instance, if

$$f = F_N + o(\phi_N), \quad g = G_M + o(\psi_M),$$

then

$$fg = F_N G_M + F_N o(\psi_M) + G_M o(\phi_N) + o(\phi_N \psi_M).$$

After the products are reordered by size, this yields the product asymptotic expansion. Inversion follows from factoring the leading term and applying a geometric expansion with a controlled remainder.

Composition and differentiation require extra hypotheses. Differentiating an o -remainder need not preserve its order, and composing can move the argument outside the uniformity region. This is why a formal identity and an analytic asymptotic theorem are logically distinct statements.

Checkpoint

Most elementary transseries calculations rest on four moves:

1. factor the dominant monomial;
2. expand only a genuinely small remainder;
3. cancel the largest residual term;
4. verify that supports remain locally finite and well ordered.

The deep theory makes these moves into closure theorems for large fields.

Glossary

The entries below favor working intuition, followed by the formal point that matters in calculations.

Action

A positive quantity A appearing in an exponential scale such as e^{-Ax} . In resurgence, Borel singularities often occur at locations related to actions.

Alien derivative

An operator, usually written Δ_A , designed to extract the resurgent information carried by a Borel singularity at $p = A$. It is not ordinary differentiation with respect to x .

Anti-Stokes direction

A direction along which two competing exponentials have equal magnitude, so dominance can exchange. Terminology varies in the literature; some authors interchange the labels “Stokes” and “anti-Stokes.”

Asymptotic couple

An ordered abelian group together with a map recording the dominant value of logarithmic derivatives. It abstracts the interaction between valuation and differentiation in an H -field.

Asymptotic expansion

A sequence of approximations whose successive remainders are smaller in a specified asymptotic scale. It need not converge for any fixed value of the variable.

Asymptotic scale

A sequence $\phi_0 \succ \phi_1 \succ \phi_2 \succ \cdots$ with $\phi_{n+1} = o(\phi_n)$ on the limit regime under study.

Beyond all orders

Smaller than every term in a chosen perturbative scale. For example, e^{-x} is beyond all orders of x^{-1} as $x \rightarrow +\infty$.

Borel transform

A transform that divides factorial growth from coefficients. In the convention of this book, $\sum_{n \geq 0} a_n x^{-n-1}$ maps to $\sum_{n \geq 0} a_n p^n / n!$.

Borel sum

An analytic function obtained by analytically continuing a Borel transform and applying a directional Laplace integral. It assigns meaning to many divergent Gevrey series.

Compositional inverse

A series G satisfying $F \circ G = G \circ F = x$. It is different from the multiplicative inverse $1/F$.

Dominance

The comparison $F \succ G$, meaning $G/F \rightarrow 0$ in an analytic setting or that the leading monomial of F is larger in a formal ordered field.

EL-series

Exponential-logarithmic series built by a construction order that starts from exponentials and then closes under logarithmic operations. Their relation to LE-series is subtle rather than merely notational.

Exponential height

A rough measure of how many nested exponential layers are needed to construct a monomial or transseries.

Flat function

A nonzero function with zero asymptotic power series in a given scale. The standard example at $+\infty$ is e^{-x} relative to inverse powers.

Formal solution

A transseries that makes an equation vanish under formal substitution. This alone does not assert that an actual analytic function realizes the transseries.

Gevrey order

A quantitative rate of coefficient growth. Gevrey-1 typically means $|a_n| \leq CA^n n!$; such growth is suited to ordinary Borel transformation.

Germ at $+\infty$

An equivalence class of functions that agree on some sufficiently far tail interval. Germs forget behavior on bounded sets.

Grid

A finitely generated multiplicative lattice of monomials, often written $\mu^{\mathbf{k}}$ for integer multi-indices $\mathbf{k}$ in a translated cone.

Grid-based transseries

A transseries whose support lies in a finite-rank grid. This setting is especially convenient for algorithms.

Hahn series

A generalized formal sum $\sum_{\mathbf{g} \in G} a_{\mathbf{g}} \mathbf{g}$ over an ordered abelian group, with support satisfying a well-order condition that makes algebra possible.

Hardy field

A field of germs at infinity closed under differentiation. Ordering is eventual ordering of functions.

Height wins

A useful informal rule: a genuinely positive exponential layer generally outruns all combinations from lower exponential height, although signs and cancellations must still be checked.

H -field

An ordered differential field satisfying axioms modeled on Hardy fields, including compatibility between order and differentiation for large elements.

Hyperasymptotics

The repeated asymptotic analysis of optimally truncated remainders, exposing successively smaller exponential scales.

Infinitesimal

An element smaller in magnitude than every positive real constant. In transseries at $+\infty$, x^{-1} and e^{-x} are infinitesimal.

Instanton sector

In physics-inspired terminology, a sector proportional to e^{-kAx} , often multiplied by powers and a perturbative series. The label need not imply literal instantons in the physical problem.

Lateral Borel sum

A Laplace integral taken just above or just below a singular direction in the Borel plane. The difference of the two lateral sums is a Stokes jump.

LE-series

Logarithmic-exponential series obtained from a base variable by iterated algebraic operations, logarithms, exponentials, and suitable generalized summation.

Leading coefficient

The coefficient multiplying the largest monomial of a nonzero transseries.

Leading monomial or magnitude

The largest monomial in the support of a nonzero transseries. It determines first-order size.

Liouville closed

An ordered differential field in which algebraic equations, antiderivatives, and logarithmic antiderivatives have the expected solutions. This is a differential-algebraic closure property, not ordinary topological completeness.

Logarithmic depth

A measure of how many iterated logarithms are needed below the main variable, as in $\log x$, $\log \log x$, and so on.

Monomial

A formal object representing one pure order of growth, such as $x^a(\log x)^b e^{-x}$.

Newton polygon

A geometric device for balancing terms and discovering candidate leading exponents. Differential versions organize dominant balances for ODEs.

Newtonian H -field

Roughly, an H -field in which differential polynomials of Newton degree one have zeros. This property plays a role analogous to algebraic closedness for differential equations.

Normal form

A canonical decomposition such as $T = a\mathfrak{m}(1 + \varepsilon)$ with $a \neq 0$, dominant monomial $\mathfrak{m}$, and $\varepsilon \prec 1$.

Omega-series

A large class of surreal transseries generated from the simplest infinite surreal number ω by transserial operations and summation.

Optimal truncation

Stopping a divergent asymptotic series near its least term. The resulting error is often exponentially small and close to the best accuracy available from that sector alone.

Ratio set

A finite set of small monomials generating a grid. Ratio sets provide a computational encoding of relative smallness.

Resonance

A degeneracy in coefficient matching, often caused by an exponent difference or linearized eigenvalue hitting an integer relation. It can force logarithms or new free parameters.

Resurgence

A property in which the Borel transform has controlled analytic continuation and its singularities encode relations among different transseries sectors.

Saddle point

A critical point of an exponent in a Laplace- or steepest-descent integral. Different saddles produce different exponential sectors.

Sector

Either a geometric angular region in the complex plane or, by context, one exponential block $e^{-kAx}\Phi_k(x)$ of a transseries.

Stokes automorphism

An operator describing how a resummed transseries changes when a Stokes direction is crossed.

Stokes constant

A problem-dependent constant measuring the strength of a Stokes connection between sectors. Its numerical normalization depends on conventions.

Stokes direction

A Borel-Laplace direction obstructed by singularities, across which lateral sums differ. In WKB language, naming conventions can vary.

Summable family

A formal family for which every monomial receives only finitely many coefficient contributions

and whose union of supports remains legal.

Support

The set of monomials with nonzero coefficient in a generalized series.

Surreal number

An element of Conway's proper-class-sized ordered field containing the real numbers, infinite and infinitesimal numbers, and a rich exponential structure.

Transmonomial

A monomial in a transseries group, including nested exponential and logarithmic scales rather than only ordinary powers.

Transseries

A generalized formal series built from ordered monomials and closed, in an appropriate framework, under operations such as exponentiation, logarithm, differentiation, integration, and composition.

Valuation

A map recording the leading order of a nonzero element. Depending on convention, a larger valuation may correspond to a smaller magnitude.

Well-based support

A support with no forbidden infinite ascent in monomial size; equivalently, under the opposite valuation order it is well ordered. This ensures a largest term and controls products and substitutions.

WKB method

A method for differential equations with a large or small parameter, using exponential phases and amplitude series. It naturally produces transseries and Stokes phenomena.

Notation and formula sheet

C.1 Basic symbols

Symbol	Meaning
$F \prec G$	$F/G \rightarrow 0$, or the corresponding formal dominance relation.
$F \asymp G$	Comparable leading size; in some analytic contexts, bounded above and below by constant multiples.
$F \sim G$	Usually $F/G \rightarrow 1$ for functions; elsewhere in the book, an explicit series after $\sim$ denotes an asymptotic expansion.
$\mathfrak{m}, \mathfrak{n}, \mathfrak{g}$	Generic transmonomials.
$\mathfrak{G}$	An ordered multiplicative group of monomials.
$\text{supp } T$	The set of monomials having nonzero coefficient in T .
$\text{mag } T$	The leading or largest monomial of T .
$\mathbb{T}$	A standard field of logarithmic-exponential transseries.
$\mathcal{B}$	Formal Borel transform.
$\mathcal{L}_\theta$	Laplace transform along the ray of argument θ .
$\mathcal{S}_\theta$	Directional Borel-Laplace summation operator.
$\mathfrak{S}_\theta$	Stokes automorphism across direction θ .
Δ_A	Alien derivative associated with a Borel singularity/action A .
C or σ	Free transseries parameter multiplying a homogeneous/exponential sector.
A	Exponential action in e^{-Ax} .
W	Lambert's inverse of $w \mapsto we^w$.
ω	The simplest positive infinite surreal number.

C.2 Canonical algebraic forms

Core idea

For a nonzero transseries,

$$T = a\mathfrak{m}(1 + \varepsilon), \quad a \neq 0, \quad \varepsilon \prec 1.$$

Then

$$T^{-1} = a^{-1}\mathfrak{m}^{-1} \sum_{n \geq 0} (-\varepsilon)^n,$$

and, for positive T ,

$$\log T = \log a + \log \mathfrak{m} + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \varepsilon^n.$$

For real α ,

$$T^\alpha = a^\alpha \mathfrak{m}^\alpha \sum_{n \geq 0} \binom{\alpha}{n} \varepsilon^n.$$

If $U = U_{\succ 1} + c + U_{\prec 1}$, then

$$e^U = e^c e^{U_{\succ 1}} \sum_{n \geq 0} \frac{U_{\prec 1}^n}{n!}.$$

For a monomial $\mathfrak{m} = \exp(L)$,

$$\mathfrak{m}' = L' \mathfrak{m}.$$

For $\mathfrak{m} = x^a (\log x)^b e^L$,

$$\mathfrak{m}' = \mathfrak{m} \left(\frac{a}{x} + \frac{b}{x \log x} + L' \right).$$

C.3 A standard one-parameter sector ansatz

For one action $A > 0$ and power shift β ,

$$y(x; C) = \sum_{k \geq 0} C^k e^{-kAx} x^{k\beta} \Phi_k(x), \quad \Phi_k(x) = \sum_{n \geq 0} a_{k,n} x^{-n}.$$

With two independent actions,

$$y(x; C_1, C_2) = \sum_{k, \ell \geq 0} C_1^k C_2^\ell e^{-(kA + \ell B)x} x^{k\alpha + \ell\beta} \Phi_{k, \ell}(x).$$

Nonlinearity adds sector indices because exponential monomials multiply.

C.4 Borel–Laplace conventions used here

For

$$\tilde{f}(x) = \sum_{n \geq 0} a_n x^{-n-1},$$

define

$$\widehat{f}(p) = \mathcal{B}\widetilde{f}(p) = \sum_{n \geq 0} \frac{a_n}{n!} p^n.$$

The directional Laplace transform is

$$\mathcal{S}_\theta \widetilde{f}(x) = \int_0^{e^{i\theta}\infty} e^{-xp} \widehat{f}(p) dp,$$

when continuation and growth permit. The basic identity

$$\int_0^{e^{i\theta}\infty} e^{-xp} p^n dp = \frac{n!}{x^{n+1}}$$

holds in a compatible sector and explains why the Laplace transform recovers the original formal powers.

Multiplication becomes convolution:

$$\mathcal{B}(\widetilde{f}\widetilde{g}) = \widehat{f} * \widehat{g}, \quad (\widehat{f} * \widehat{g})(p) = \int_0^p \widehat{f}(q) \widehat{g}(p-q) dq.$$

A simple Borel singularity on the ray produces a lateral jump proportional to e^{-Ax} .

C.5 Frequently used model expansions

Lambert W at $+\infty$

With $L_1 = \log x$ and $L_2 = \log \log x$,

$$\begin{aligned} W(x) = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2^2/2 - L_2}{L_1^2} \\ + \frac{L_2^3/3 - (3/2)L_2^2 + L_2}{L_1^3} + \mathcal{O}\left(\frac{L_2^4}{L_1^4}\right). \end{aligned}$$

Airy decay

For the recessive solution of $y'' = xy$,

$$\text{Ai}(x) \sim \frac{1}{2\sqrt{\pi}} x^{-1/4} e^{-2x^{3/2}/3} \left(1 - \frac{5}{48x^{3/2}} + \frac{385}{4608x^3} - \dots \right).$$

Stirling expansion

As $x \rightarrow +\infty$,

$$\log \Gamma(x) \sim \left(x - \frac{1}{2}\right) \log x - x + \frac{1}{2} \log(2\pi) + \sum_{n \geq 1} \frac{B_{2n}}{2n(2n-1)x^{2n-1}}.$$

Euler's factorial series

The divergent series

$$\tilde{F}(x) = \sum_{n \geq 0} (-1)^n n! x^{-n-1}$$

has Borel transform $1/(1+p)$ and Borel sum

$$F(x) = \int_0^\infty \frac{e^{-xp}}{1+p} dp = e^x E_1(x)$$

for $\Re x > 0$, where $E_1(x) = \int_x^\infty e^{-t} t^{-1} dt$.

Fractional iteration through an Abel coordinate

If $V(f(x)) = V(x) + 1$, then the formal s -th iterate is

$$f^{[s]}(x) = V^{-1}(V(x) + s).$$

This automatically gives

$$f^{[s]} \circ f^{[t]} = f^{[s+t]}, \quad f^{[0]} = \text{id}, \quad f^{[1]} = f.$$

C.6 A diagnostic checklist

When confronted with a new asymptotic problem, ask:

1. What is the limit regime, and which quantities are large or small?
2. What dominant balance selects the first monomial?
3. Which ratios are genuinely small enough to expand?
4. Does the equation possess a homogeneous solution invisible to the perturbative scale?
5. Does nonlinearity force multiples or combinations of exponential actions?
6. Are logarithms required by resonance?
7. Is the coefficient growth factorial, and where are the predicted Borel singularities?
8. On which complex sectors should an actual solution realize the formal object?
9. What changes on crossing a Stokes direction?
10. Which claims are formal identities, and which need analytic estimates?

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